

Digital Image Processing: Introduction

***“One picture is worth more
than ten thousand words”***

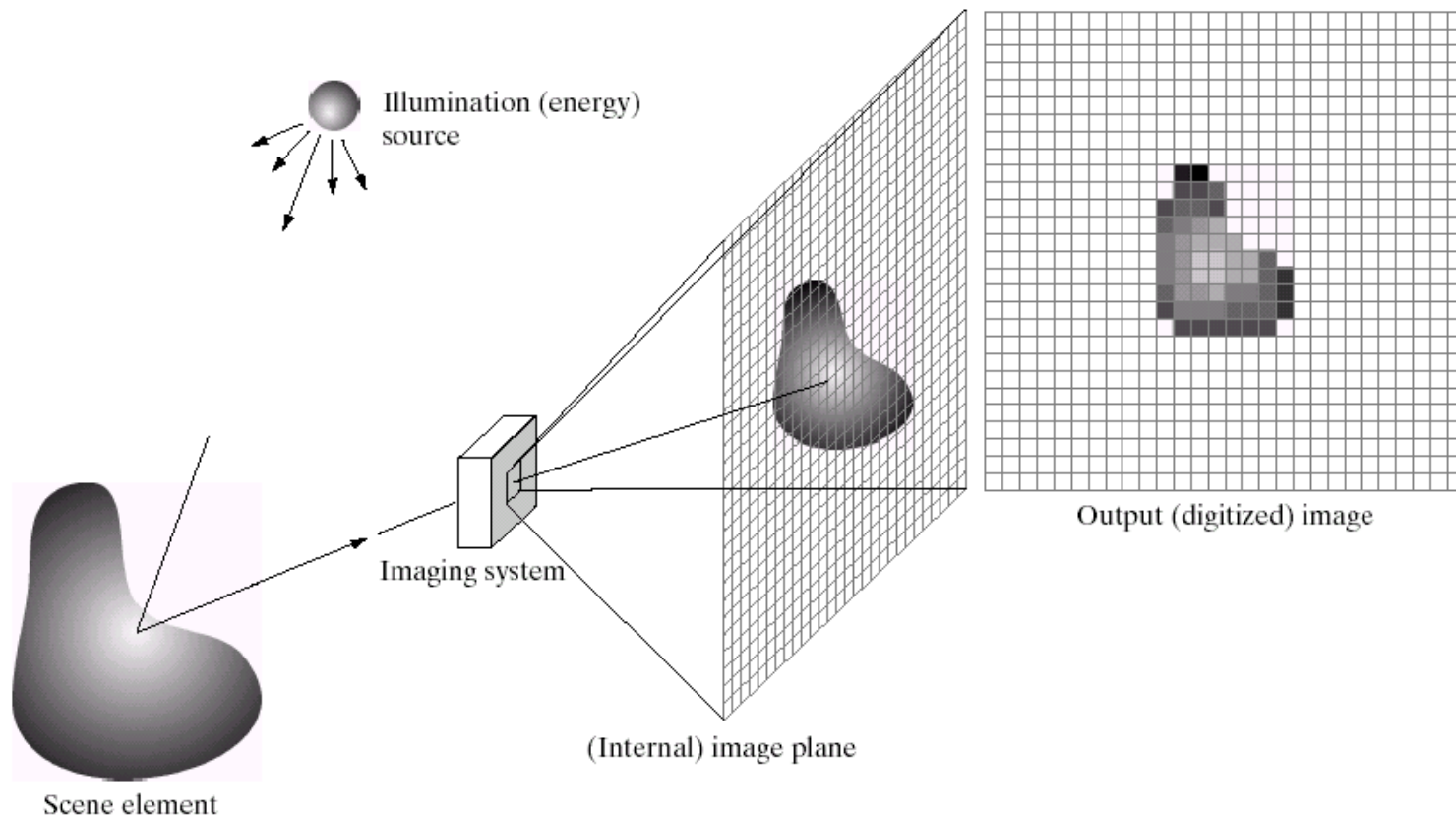
What is an Image?

- Image is a representation of something or someone. Examples are any drawing, painting, photograph etc.
- Images are very powerful tool in communication.
- It may be black & white or color.



What is a Digital Image?

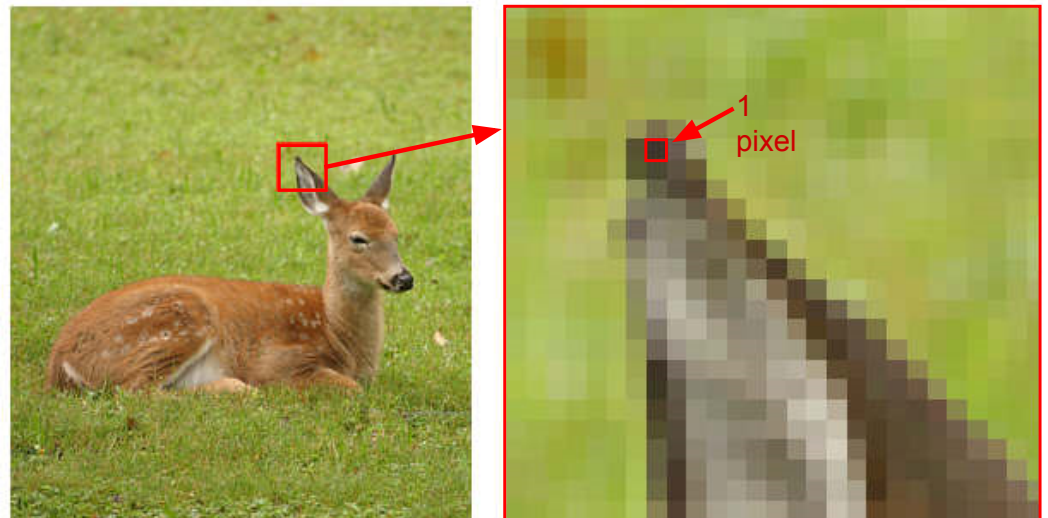
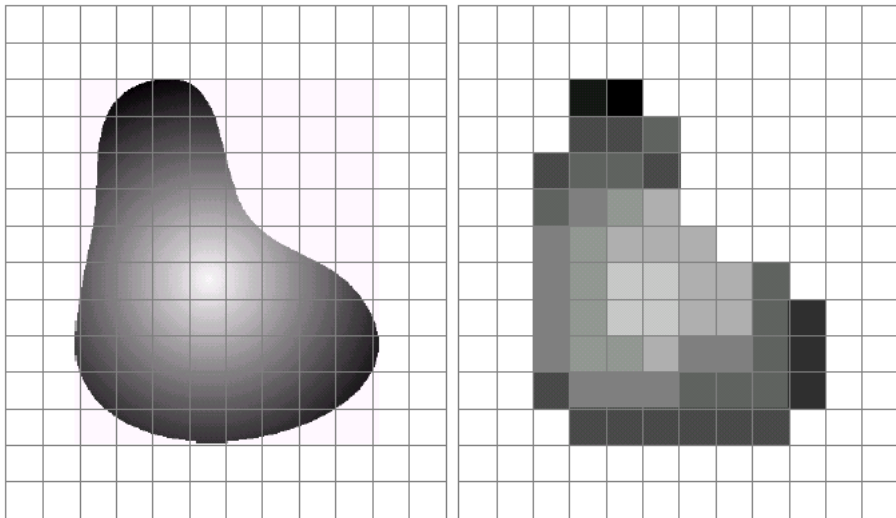
A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels



What is a Digital Image? (cont...)

Pixel values typically represent gray levels, colours, heights, opacities etc

Remember *digitization* implies that a digital image is an *approximation* of a real scene



What is digital Image?

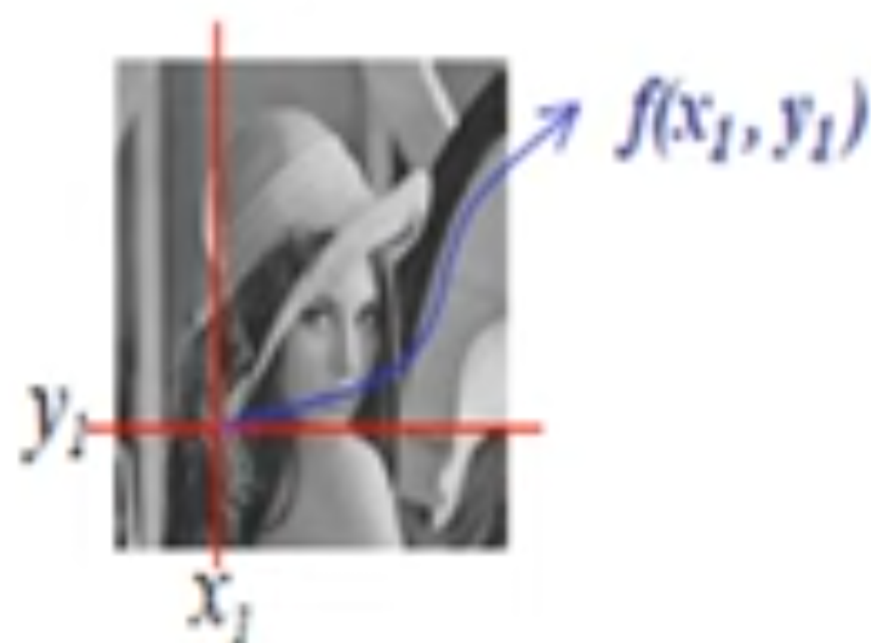
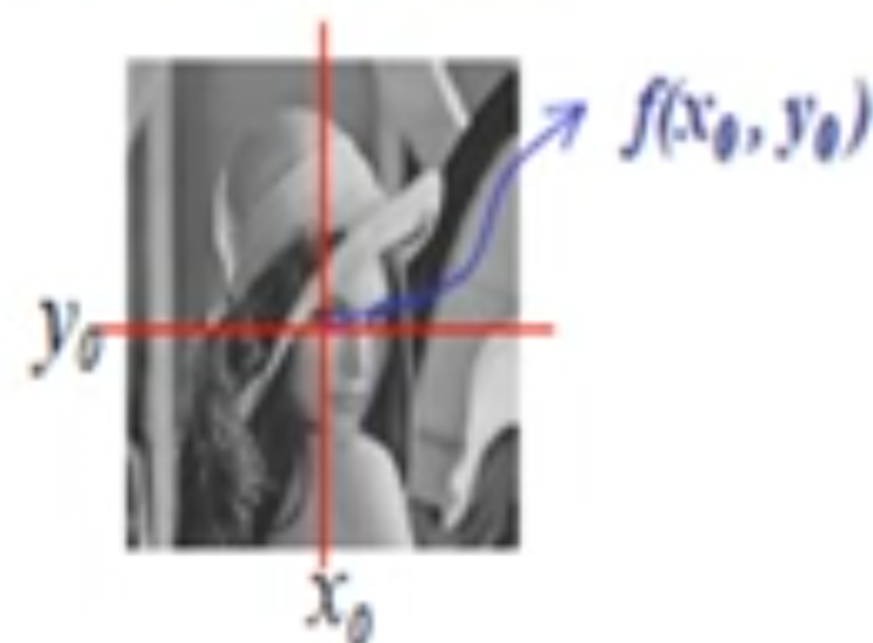
- When x , y , and $f(x, y)$ are all finite, discrete quantities, we call the image a **digital image**.
- Digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are referred to as:
 - picture elements
 - image elements
 - Pels
 - **pixels**
- Pixel is the most widely used term.



Often an image is defined as a two-dimensional function, $f(x, y)$.

➤ where x and y are spatial (plane) coordinates

➤ the amplitude of f at any pair of coordinates (x, y) is called the **intensity or gray level** of the image at that point.



Types of Images

Classification of Images on the basis of **Attributes:**

- **Vector** files are most useful when a graphic needs to be able to handle resizing, as they retain crisp detail when scaled to any size. Vector format is an excellent choice for digital illustrations such as logos.
- A vector file *can* be converted to a raster file, but it will permanently lose the advantages of the vector format.
- **Common vector file extensions: SVG, EPS, and EMF**
- SVG — web-based vector format
- EPS — Adobe-based vector format
- EMF — Microsoft Office-based vector format

- **Raster**
- **Raster** images are best for complex images like photographs, and are more likely to be a widely compatible file format (such as JPG). However, raster images lose detail and become increasingly blurry when enlarged.
- Although a vector image can be saved as a raster image, it is *not possible* to convert a raster image into a vector image. Raster files saved as vector file formats will still be raster files. Most images on the web are raster files.

- **Common raster file extensions: JPG, PNG, and TIFF**
- JPG — compressed raster format, often used for photos
 - Best for web use
 - Small file size
 - Universal — compatible with most software
 - Most images on stock photography websites will be downloaded as JPG files
- PNG — raster format good for illustrations or icons
 - supports transparency, unlike JPG
 - Best for web use
 - Small file size

- TIFF — raster format that is best for high-resolution printing
 - Great for professional print use, especially when preserving crisp image detail is important
 - Large file size

- Classification of images on the basis of **colour**: On the basis of colour, the images can be classified into the following categories:
- Monochrome images are the images where the colour component is absent, they are further classified as Grey scale and binary images.
 - o Grey scale images : The term grey scale refers to the range of shades between white and black or vice versa.

- such images have many shades of grey, and eight bits ($2^8 = 256$) are enough to represent the grey scale because the human visual system can not distinguish more than 32 different grey levels, and the additional bits are necessary to cover noise margins. Most medical images such as X-rays, CT images, MRIs, and ultrasound images are grey scale images.

- Binary images :The binary images are just special case of grey scale images, where the process of Thresholding is applied, they are actually Bi-level images where the pixels assumes the values of 0 or 1, i.e. only one bit is sufficient to represent the pixel value.

Digital Image Processing

- Digital
 - Image
 - Processing

Processing of Image which are digital in nature
by a Digital Computer.

Motivation....

- Improvement of pictorial information
- Efficient storage & transmission etc.



What is Image Processing?

- Image processing is a method to convert an image into digital form and perform some operations on it, in order to get an enhanced image or to extract some useful information from it.
- It is a type of **signal dispensation** in which input is image, and output may be image or characteristics associated with that image.
- It is among rapidly growing technologies today, with its applications in various aspects of a business. Image Processing forms core research area within engineering and computer science disciplines too.



What is Digital Image Processing?

Digital image processing focuses on two major tasks

- Improvement of pictorial information for human interpretation
- Processing of image data for storage, transmission and representation for autonomous machine perception

Some argument about where image processing ends and fields such as image analysis and computer vision start

What is DIP? (cont...)

The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes

Low Level Process	Mid Level Process	High Level Process
Input: Image Output: Image	Input: Image Output: Attributes	Input: Attributes Output: Understanding
Examples: Noise removal, image sharpening	Examples: Object recognition, segmentation	Examples: Scene understanding, autonomous navigation

Purpose of Image processing

The purpose of image processing is divided into 5 groups. They are:

- **Visualization** - Observe the objects that are not visible.
- **Image sharpening and restoration** - To create a better image.
- **Image retrieval** - Seek for the image of interest.
- **Measurement of pattern** – Measures various objects in an image.
- **Image Recognition** – Distinguish the objects in an image.



Methods used for Image Processing

Two methods are used for Image Processing are:

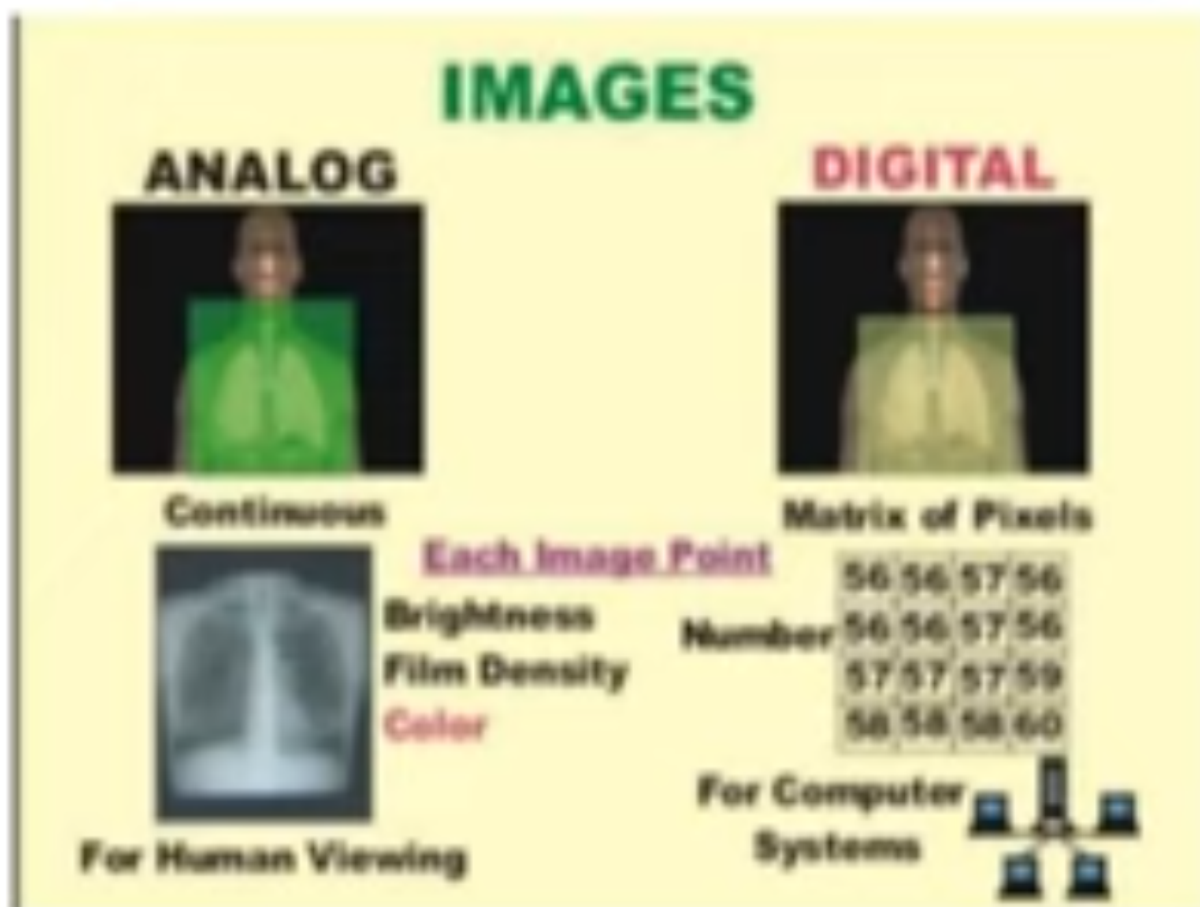
Analog image processing: In this type of processing, the images are manipulated by electrical means by varying the electrical signal. The common example include is the television image.

Digital image processing: It help in manipulation of the digital images by using computers. The three general phases that all types of data have to undergo while using digital technique are:

- Pre- processing
- Enhancement and display
- Information extraction



Example: Let 1 represents white and 0 represents black color.



Digitization of Image

- Image digitization refers to the process of sampling and quantization of the analog signals, which is required to convert the analog signal in to digital form. The digital image processing is performed using Digitizers, and this digitization process consists of two steps. One is called sampling and the other is called quantization.

- Sampling refers to considering the image only at a finite number of points, each image sample is called a pixel and
- Quantization refers to the representation of the grey level value at the sampling point using finite number of bits. For example 256 level quantization results in image storage of 8 bits per pixel.

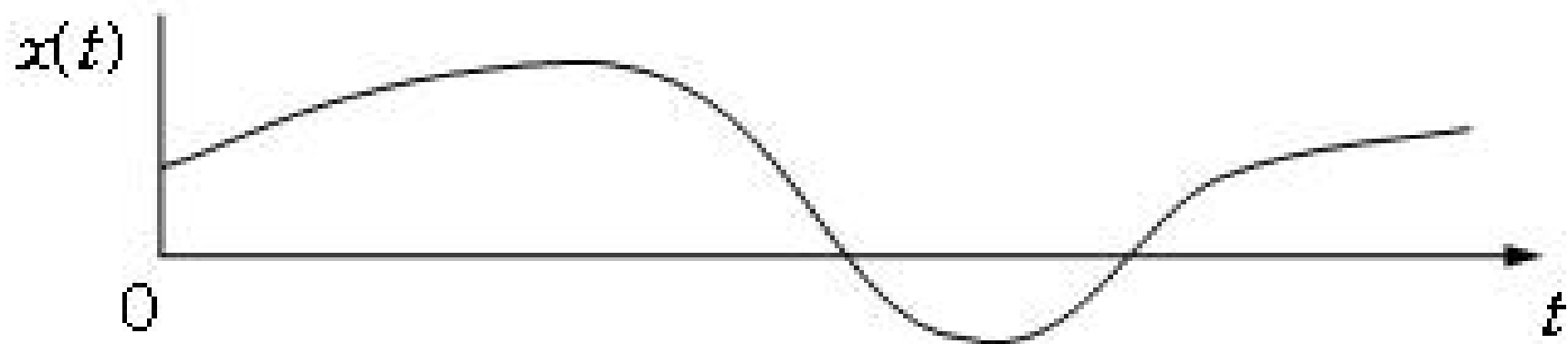
What is an analog image?

- When we capture the image of an object, we use image sensors to sense the incoming light and form the image.
- Image sensors convert the incoming light from an object into electrical signals that can be stored and viewed later.
- These analog signals are continuous. The images are stored in an analog form.

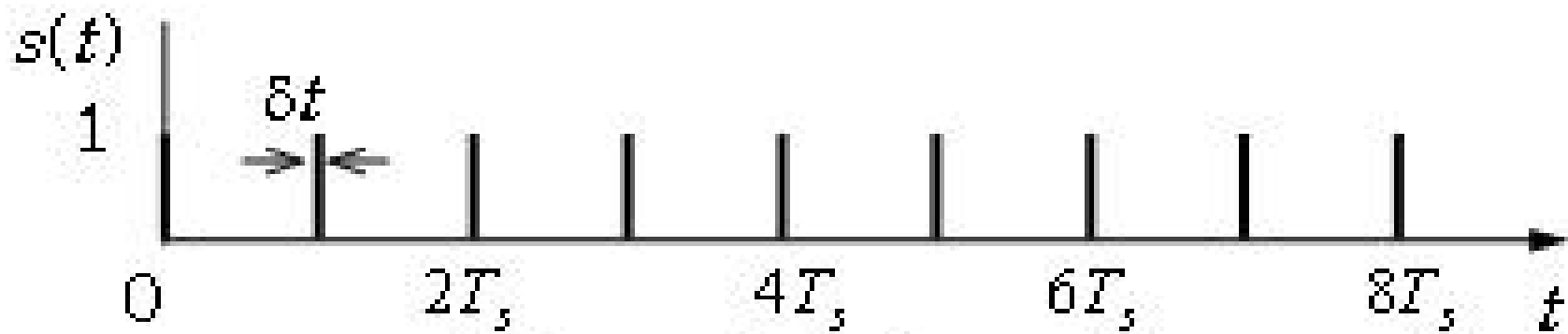
- We cannot process analog images by a computer.
- Analog signals contain infinite points, and we need infinite memory to store them.
- We need to convert the analog images into digital images to store and process by a computer.

- **Analog image to digital image conversion**
- An analog image is converted to a digital image by digitizing the analog signals. We apply sampling and quantization to the analog signals to convert them into digital form.
- A digital image is formed by arranging pixels in rows and columns. Each pixel has a particular integral value. The computer process that integral value and show us that pixel, the arrangement of the pixels form the digital image.

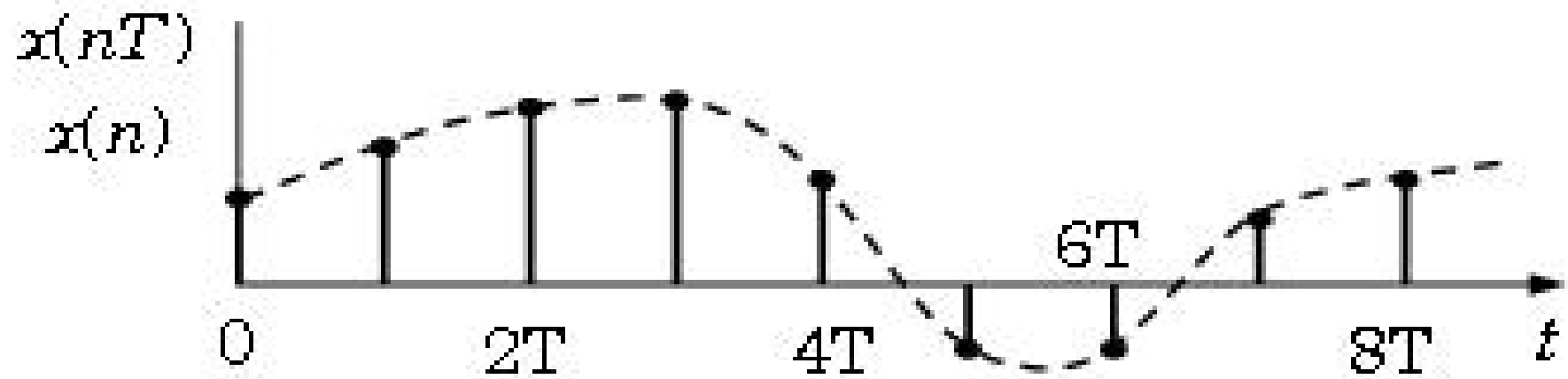
- To create a digital image, we need to convert the continuous sensed data into digital form. This process includes 2 processes:
- **Sampling:** Digitizing the co-ordinate value is called sampling.
- **Quantization:** Digitizing the amplitude value is called quantization.
- To convert a continuous image $f(x, y)$ into digital form, we have to sample the function in both co-ordinates and amplitude.



(a) Analog signal



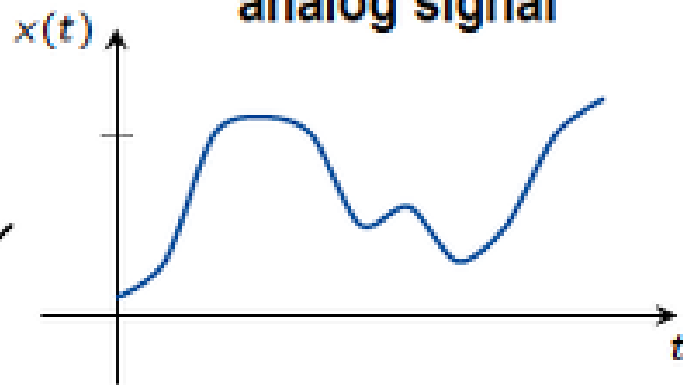
(b) Sampling function



(c) The samples (discrete-time signal)

- **Image quantization**
- After sampling the analog signal, we will apply quantization. Quantization digitizes the amplitude of the sampled signal. Quantization is done by rounding off the amplitude of each sample and then assigning a different value according to its amplitude. Each value will represent a different color tone.

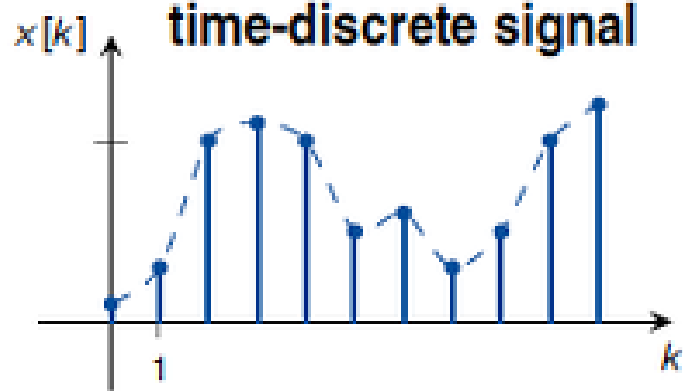
analog signal



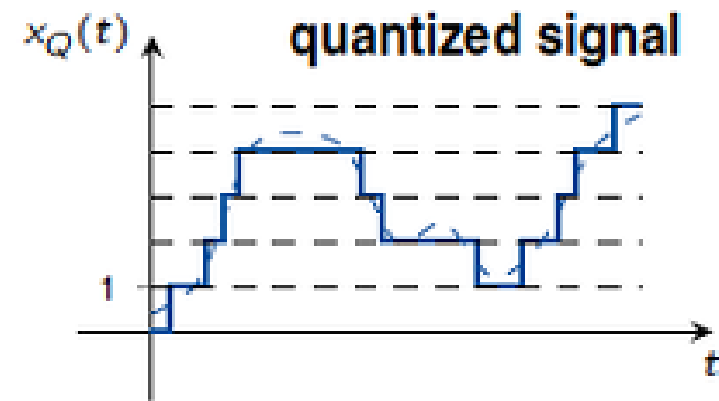
sampling

quantization

time-discrete signal



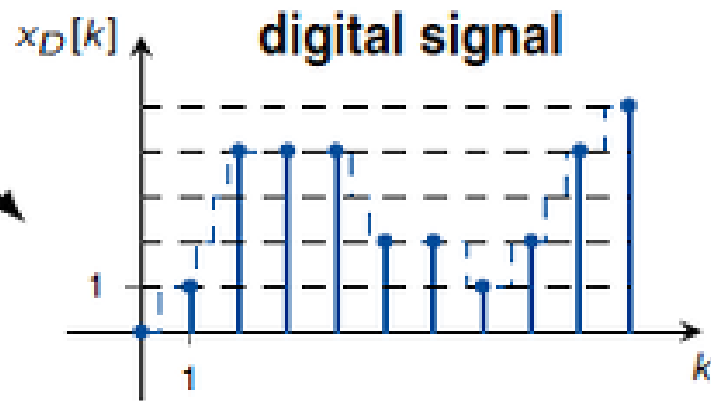
quantized signal



quantization

sampling

digital signal



Sampling

Quantization

Determines the number of pixels of a digital image.

Determines the tone of a pixel

Sampling digitizes the x-axis of an analog signal.

Quantization digitizes the y-axis of an analog signal.

The amplitude value of the analog signal is recorded at regular time intervals.

The amplitude values are rounded off and given a value.

Done prior to quantization.

Done after sampling.

Pixelization error

- The size of the pixel is important for image quality. If the size is very large, there will be a lesser number of pixels. Hence, the details become less, which makes the image distracting and meaningless. This is called **pixelization** error where the grey level discontinuities at the edges of the pixel become poor

- Then question arises that, What should be the ideal size of the pixel? Should it be big or small?
- The answer is give by the Shannon-Nyquist theorem. As per Shannon-Nyquist theorem, the sampling frequency should be greater than or equal to $2 \times f_{\max}$ where $f_{\max}$ is the highest frequency present in the image. Otherwise, the original signal cannot be reconstructed.

Example1

- Example 1: What should be the physical size of a 2D image of a document with dimensions is 2400X 2400 , when scanned at 300dpi. Here dpi stands for dots per inch.
- **Solution:** The physical size of image is
$$=(\text{Number of pixels in width}/\text{Resolution}) \times (\text{Number of pixels in Height}/\text{Resolution})$$
$$=(2400/300) \times (2400/300)$$
$$= 8 \text{ inches} \times 8 \text{ inches}$$

Example-2

- If the physical size of a medical image is 8×8 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better quality image? Will an image of size 256×256 be enough?
- Solution: It is given that the sampling resolution is 5 cycles/mm. Therefore, for better quality, 2 pixels per cycle are required.

- This is because sampling theorem states that the sampling frequency should be greater than twice the maximum signal frequency. This means that 10 pixels per mm are required. So, the pixel size is 0.1 mm.

It may note here that this double rates in both directions are called Nyquist rates. The given image size (since 1inch = 2.54cm) is 8 X2.54X 8 X2.54 sq.cm =20.32 X 20.32 sq.cm

Therefore, the minimum number of required pixels is 2032X 2032 pixels. So, an image of size 256X 256is not enough.

Image resolution

- Image resolution describes the image's level of detail – higher resolution means more image detail. In digital imaging, the resolution is often measured as a pixel count.
- A pixel (short for picture element) is a single point or a tiny square in a graphic image stored in an ordered rectangular grid.
- It is the smallest element in a digital image.

IMAGE CHARACTERISTICS

- **Intensity**: The term Intensity refers to the amount of light or the numerical value of a pixel, it is the measure of energy of a wave, which is directly proportional to square of amplitude of the signal.
- From the point of view of image processing it is a numerical value which represents a pixel, for example for an 8-bit gray scale image the value of a pixel can be in the range of $[0, 255]$, where (say) numeric value “0” represents black colour and the numeric value “255” represents white colour.

- Therefore, the higher the value of a pixel(i. e. intensity) whiter the pixel will appear, i. e. in a grey scale images, intensity is depicted by the grey level value at each pixel i.e., two pixels with grey level value 127 and 220, then it can be interpreted that pixel with value 127 is darker than pixel with value 220 .

- **Contrast:** The term Contrast of an image relates to recording of the differences in the magnitude of the intensity at the surface of an object.
- There are many ways in which contrast can be measured, i.e. it can be described as a product of the sensor signal contrast (this depends on the energy source and the physical properties of the object) and detector contrast (this depends on the way signal is detected, captured, and stored)

- **Brightness:** It refers to the average pixel intensity of the image, Similar to contrast, the image should be reasonably bright to show all the information to the viewer.
- However, excessive brightness may affect the quality of the image. An image may have a higher brightness but if the intensity is not optimal, the image again will not exhibit good quality. Brightness resolution is defined as the number of identified distinct colours and it is also called as colour resolution.

- **Noise:** Noise is an unwanted disturbance that causes fluctuations in the pixel value. It is a random or stochastic process and hence its true value cannot be predicted accurately. However, noise obeys all the statistic properties.

What is a signal ?

- Any physical phenomenon that conveys or carries some information can be called a signal. A signal is anything you can see, hear, observe or measure using some machine.
- For examples: Speech, audio, light, radio, TV, radar, supersonic, temperature, ECG, EEG, etc .
- Usually, the information carried by a signal will be a function of an independent variable.
- The independent variable can be time, spatial coordinates, intensity of colors, pressure, temperature, etc.
- The most popular independent variable in signal is time and it is represented by the letter “t”.

- Depending on the number of dependent variables:
- One dimensional signals
- Multidimensional signals
- **One Dimensional Signals**
- A signal which is a function of single independent variable is called one dimensional signal.
- Examples: music, speech, heart beat, etc.

- **Multidimensional Signals**
- A signal which is a function of two or more independent variables is called multidimensional signal.
- Examples: A photograph is an example of two dimensional signal, the motion picture of a black and white TV is an example of three dimensional signal.

Definition(1-D Signal): (Mathematically)

- A 1-dimesional (1-D) signal is a function of a single variable. A one-dimensional continuous signal is represented as a function, $f(t)$, of the independent variable t , representing the evolution of a physical phenomenon across time 't'.
- For example, a 1-D audio signal, such as the electrical signal at the output of a microphone is a function of time.

Definition (2-D Signals):

- A 2-dimensional (2 – D) signal $f(x, y)$, is a function of two independent variables, x and y , such that (x, y) indicate a point in the 2D space.
- For example, if the function f shows the intensity in a still image, then $f(x, y)$ represents the value of the intensity for the pixel at the location (x, y) .

IMAGE AS A 2-D SIGNAL

- A grayscale image is a discrete 2-D signal $f(x, y)$, having two independent variables, x and y , such that $f(x, y)$ is the value of the signal at a pixel whose location in the image is given by integers x and y . The 2-D signals can be of various types such as separable, periodic, etc.
- Therefore, the 2D signal represents a digital image, where (x, y) is the location of the pixel and $f(x, y)$ is the value at this pixel.

TRANSFORMATIONS OF 2 D SIGNALS

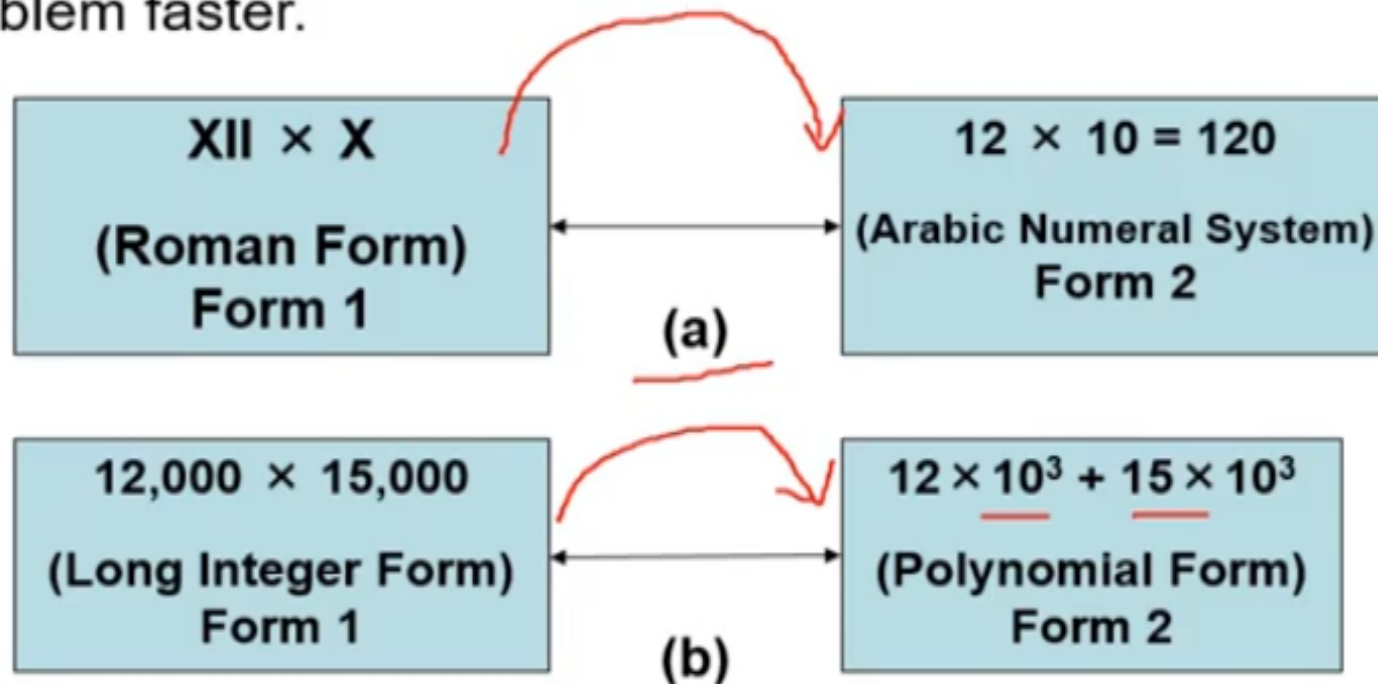
- Image transforms are necessary for image analysis and image processing. Transformations are mathematical functions that allow us to convert from one domain to another.
- Therefore, transformation of a signal $f(x)$ will convert the signal into a new signal, say, $g(y)$, in a different domain. However, transformation does not change the information content present in the signal/image.

Need for Image Transforms

Image processing applications can be applied both in:

- Spatial Domain
- Frequency Domain

A change in problem representation can help to solve the problem faster.



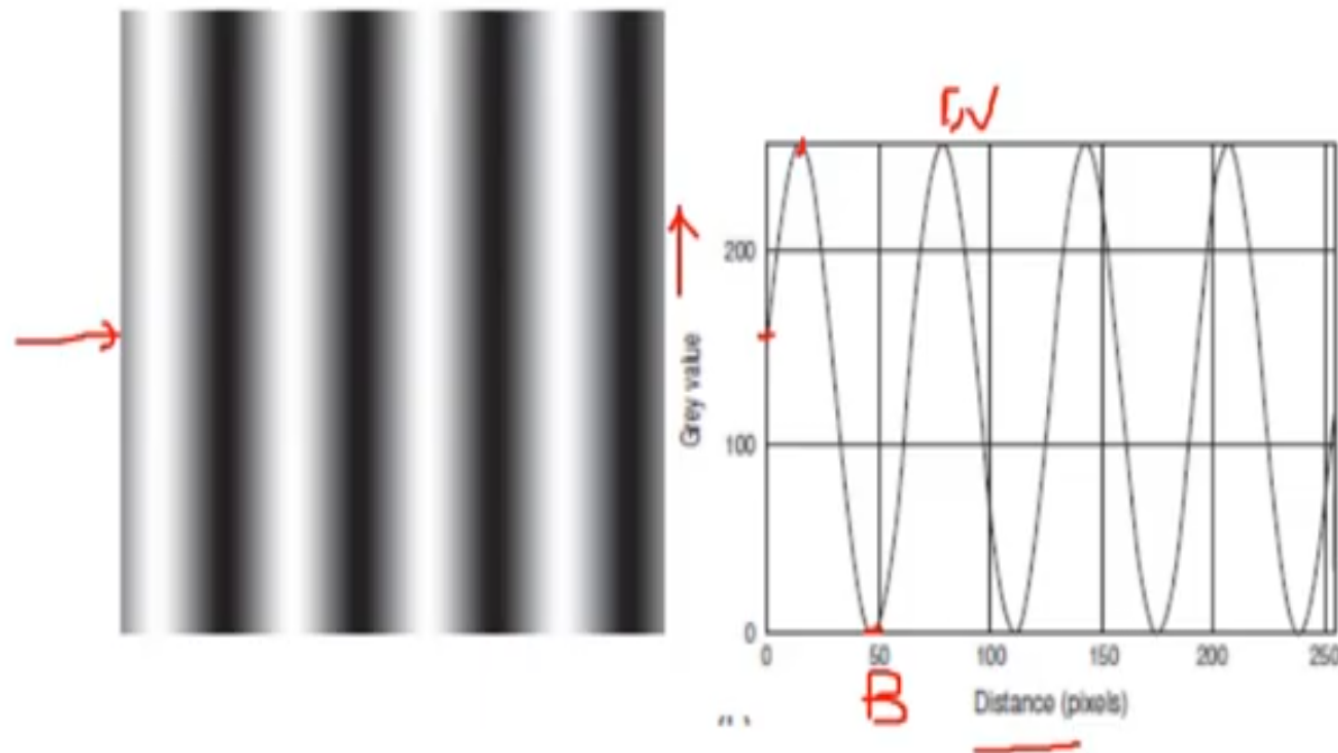


Image Profile

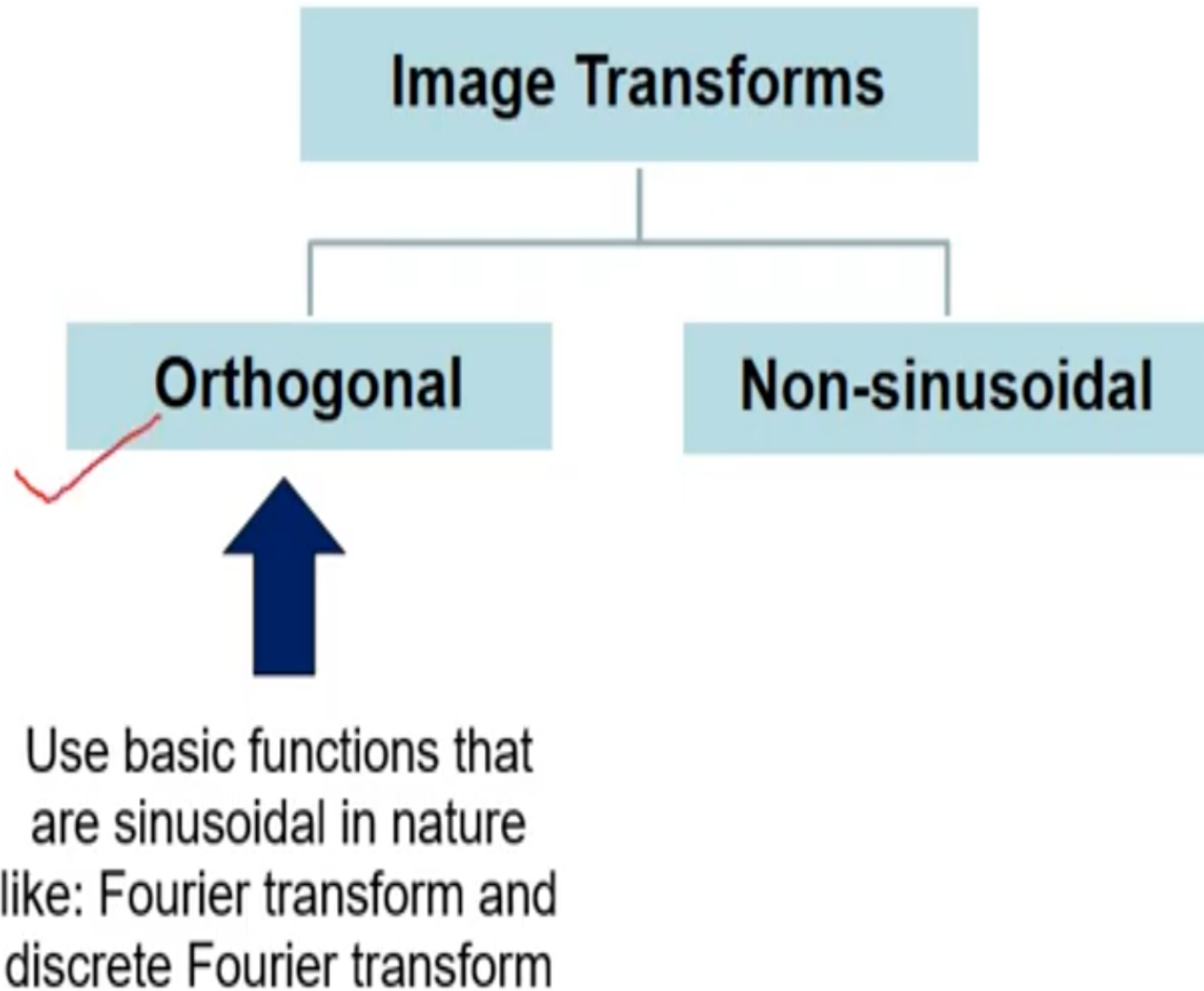
Intensity profiles are very useful in image processing.

Spatial Frequencies in Image Processing

- Term **Frequency** is used for the rate of repetition of signals in spatial domain.
- **Spatial Frequency**' is used to describe the rate of change of pixel intensity of an image in space.
- It can be visualized using the line profile of the image in any row or column.
- X-axis: distance of the pixels
- Y-axis: grey value (range from 0-255)

$$f = \frac{1}{T}$$





Basis for Transforms

An image is 2D signal so 2D transforms are required to represent image in terms of discrete set of basis images.

For $N \times N$ image of $f(x,y)$:

The forward & inverse transforms are as follows:

$$g(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u,v,x,y) \times f(x,y)$$

$$f(x,y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} I(x,y,u,v) \times g(u,v)$$



Here $T(u, v, x, y)$ and $I(x, y, u, v)$ are known as forward and inverse transformation kernels respectively.

Transformations have some important properties

Kernel is called seperable if:

$$T(u, v, x, y) = T_1(u, x) \times T_2(v, y)$$

Kernel is called symmetric if:

$$T(u, v, x, y) = T_2(u, x) \times T_1(v, y)$$



(1) Prove that Hadamard transform works for the mentioned image:-

$$\underline{F = \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}}$$

Solution → The forward transform is given as-

$$\underline{G = A \times F \times A} \quad \text{where } \underline{A = \frac{1}{\sqrt{2}} (H_{2,2})} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$G = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow \underline{G = \frac{1}{2} \begin{bmatrix} 7 & 1 \\ 1 & -1 \end{bmatrix}}$$

Inverse transform → $\underline{F = A^{-1} G A^{-1}}$

$$\underline{A^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}$$

$$F = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 7 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{2} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$F = \frac{1}{4} \begin{bmatrix} 8 & 0 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow \underline{F = \begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}}$$

IMAGE ENHANCEMENT

- Image enhancement techniques improve the quality of image as perceived by human observer/ machine vision system.
- Enhancement techniques improve the perception of information in an image for human viewing and provide 'better' input for other automated image processing techniques.
- The main objective is to modify attributes of an image to make it more suitable for a given task or a specific observer

- Image enhancement techniques are application specific and produce a 'better' image.
- They are broadly classified into two categories namely i) Spatial domain methods ii) Frequency domain methods
- In spatial (time) domain methods, pixel values are manipulated directly to get an enhanced image, **whereas** in frequency domain methods, firstly fourier transform of image is taken to convert image into frequency domain.
- Then the fourier transform is manipulated and the modified spectrum is transformed back to spatial domain to view the enhanced image.

POINT OPERATIONS

- **spatial domain enhancement techniques-**

The pixel values (grey values) of an image are directly manipulated. Such operations simply take the grey value of each pixel/neighbouring pixels, map it to a new value and move on to the next pixel.

- Let $f(x, y)$ be an input image. An image processing operation in the spatial domain may be expressed as a mathematical function $T[f(x, y)]$ applied to the image to produce a new image $g(x, y)$.

Therefore, $g(x, y) = T[f(x, y)]$

CONTRAST STRETCHING

- Contrast 'c' between two levels x_1 and x_2 are defined as the absolute values of the difference, between x_1 and x_2
i.e. $c(x_1, x_2) = |x_1 - x_2|$.
- Good contrast in an image is important to distinguish the details else they will merge in background.
- Low contrast images occur due to bad or non-uniform illumination conditions or due to nonlinearity of image acquisition devices.

- The main idea is to reduce the low level gray values, and to increase the mid range gray values, so that an artificial contrast is created between the two sets of gray values. The contrast between gray values is thus stretched from both sides.

The transformation is defined mathematically as follows:

$$s = \begin{cases} \alpha r & 0 \leq r \leq a \\ \beta(r - a) + s_a & a \leq r \leq b \\ \gamma(r - b) + s_b & b \leq r \leq L \end{cases}$$

Where α, β, γ are slopes of different regions as shown in Fig. 5 (a). The slopes determine the amount of contrast stretching (or diminishing). If the slope is greater than one, then corresponding grey levels are stretched because the original grey levels are mapped onto a larger range of grey values. A slope smaller than one means the contrast is diminished. A slope of one indicates no contrast alteration. The parameters a and b are user defined. Fig. 6 shows an example of image enhancement using contrast stretching.

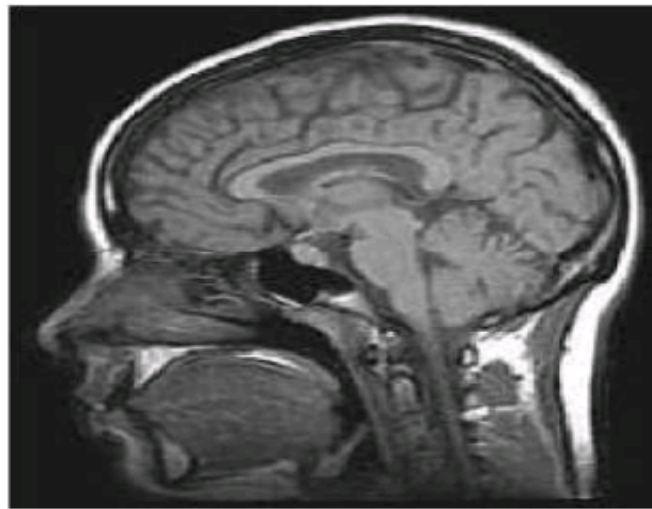
a) Clipping

If $\alpha = \gamma = 0$, this is called '**clipping**' (windowing) as shown in, Fig. 5 (b). This operation stretches the contrast to its maximum in a limited range ('window') of original grey level values $\{a, \dots, b\}$. All the grey levels outside this window are either mapped to zero or the maximum values. Thus, clipping allows us to focus all the available contrast onto the required range of grey level values. This is especially useful when viewing medical images such as CT images. It also helps in viewing under or over exposed images.

b) Thresholding

If $\alpha = \gamma = 0$ and $a = b = t$, this remapping is called '**thresholding**' as shown in Fig. 5 (c). This operation 'binarizes' (only two values present) the image. A suitable threshold value 't' is chosen, all the gray levels smaller than 't' are mapped to zero whereas all the grey levels greater than or equal to 't' are mapped to maximum value. This is a very useful operation in image processing applications. Binarization is generally done before segmentation or

region extraction. Fig. 7 shows an example of thresholding with $t = 110$. Note that input image is a gray scale image and output image is binary.



(a) original image



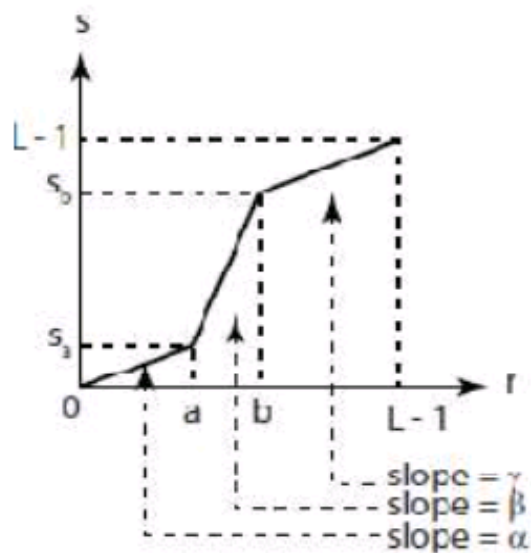
(b) image after thresholding

Fig. 7: Effect of Thresholding

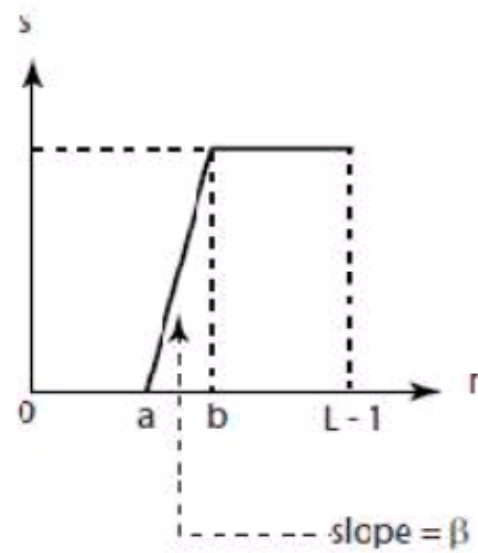
Example 3: Perform thresholding on image segment $f(x, y)$ with $t = 128$.

Solution:

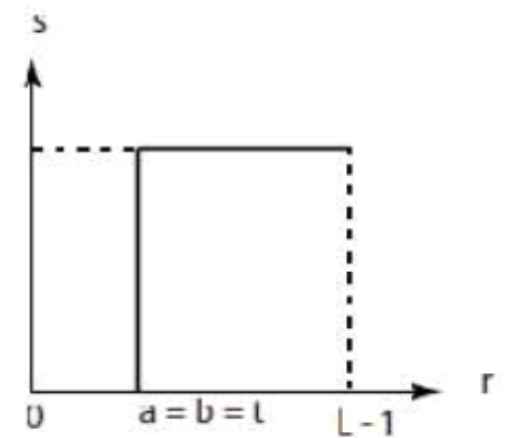
0	10	50	100	$\xRightarrow{t=128}$	0	0	0	0
5	95	150	200		0	0	255	255
110	150	190	210		0	255	255	255
175	210	255	100		255	255	255	255



(a) Contrast stretching



(b) Clipping



(c) Thresholding

Fig. 5



(a) low contrast image



(b) enhanced image

Fig. 6: Effect of Contrast Stretching

DIGITAL NEGATIVES

- The digital negative of an image with the intensity levels in the range $[0, L-1]$ is obtained by using negative transformation which is given by the expression

$$s = (L - 1) - r$$

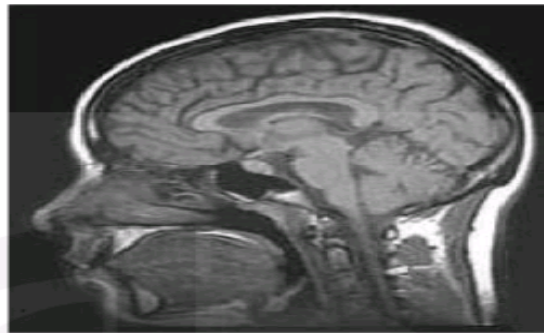
In this transformation, highest grey level is mapped to lowest and vice versa. For an 8-bit image, the transformation is

$$s = 255 - r$$

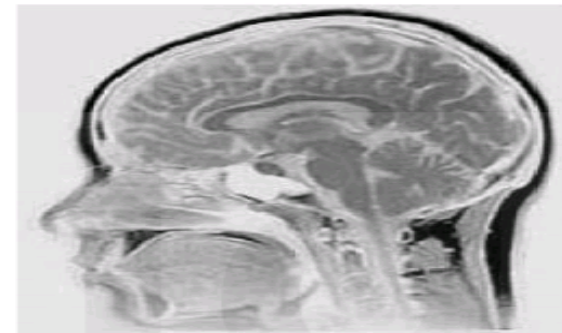
If $r = 20$, (dark pixel) $s = 255 - 20 = 235$ (bright pixel)

If $r = 125$, $s = 255 - 125 = 130$

Note that middle grey level has not changed much whereas dark grey level has become bright. This transformation is generally used to enhance the white details embedded in the dark regions of an image where black is dominant. It is useful in displaying medical images. Fig. shows an example. Even though the visual contents of both images are same, it is much easier to analyse image in negative form.



(a) original image



(b) digital negative of fig (a)

Fig. 9: Effect of digital negative

Example 4: Find the image negative transformation on an image f given as

$$f(x, y) =$$

0	10	50	100
5	95	150	200
110	150	190	210
175	210	255	100

Solution: The negative of the image is given as

$$g(x, y) =$$

255	245	205	155
250	160	105	55
145	105	165	45
80	45	0	155

INTENSITY LEVEL SLICING

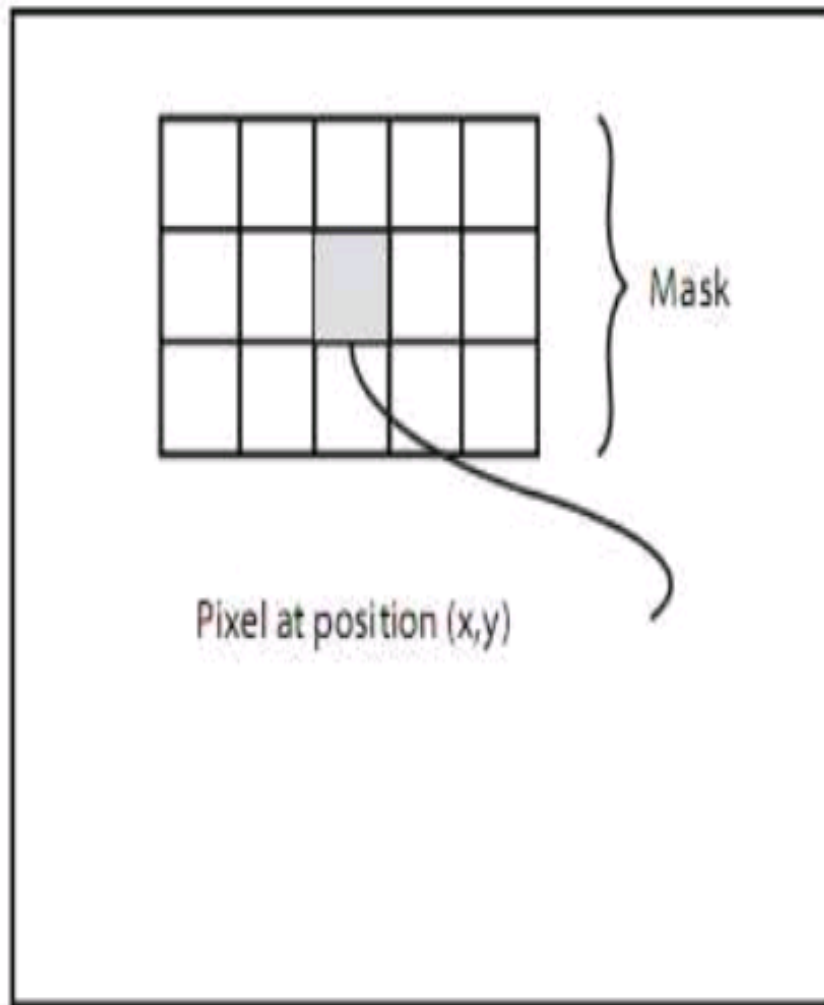
- A variant of thresholding is intensity level slicing or double thresholding. It is used to highlight a range of intensity values of interest in an image.
- In this operation, all the grey levels in a window $\{a, \dots, b\}$ are set to maximum grey level and all the grey levels outside this range are set to zero.

- Intensity level slicing is a binarization operation as the resulting image has only two grey level values (0 and M).
- Mostly this technique is used to remove unwanted elements (clutter) from an image so that the useful information becomes prominent.
- Applications are enhancing certain features which are based on grey level values, such as mass of water in satellite images, cancer cells in a monogram images etc.

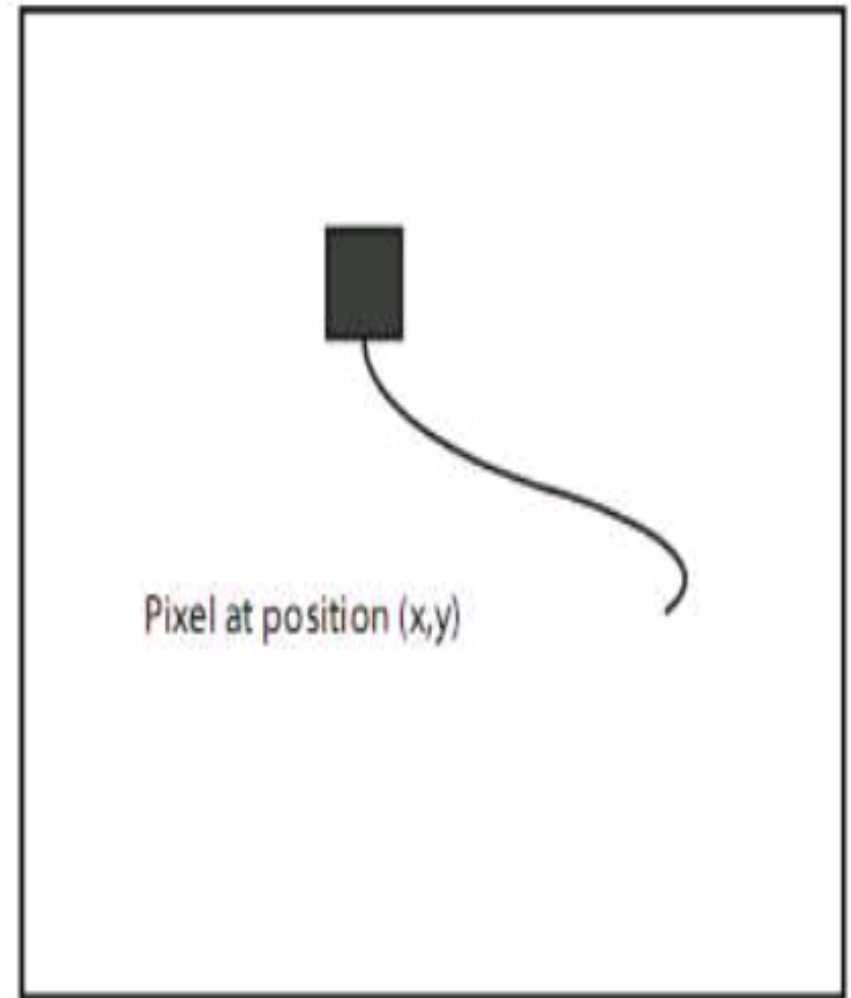
SPATIAL FILTERING

- **Filtering** is a process that removes some unwanted components or small details in an image.
- In digital image processing, filter is basically a subimage and is known by various names such as mask, kernel, template or window.
- Filters can be of two types: Spatial filters and frequency domain filters.

- **Spatial filtering** is one of the main tools used in variety of applications such as noise removal, bridging the gaps in object boundaries, sharpening of edges etc.
- The idea of spatial filtering is to move a ‘mask’, a square of odd size (such as 3x3, 5x5 etc.) over the whole image pixel by pixel.
- At each pixel, the corresponding value of the mask and the image are multiplied and added up to replace the original grey value of the pixel. This process is known as “convolution”.



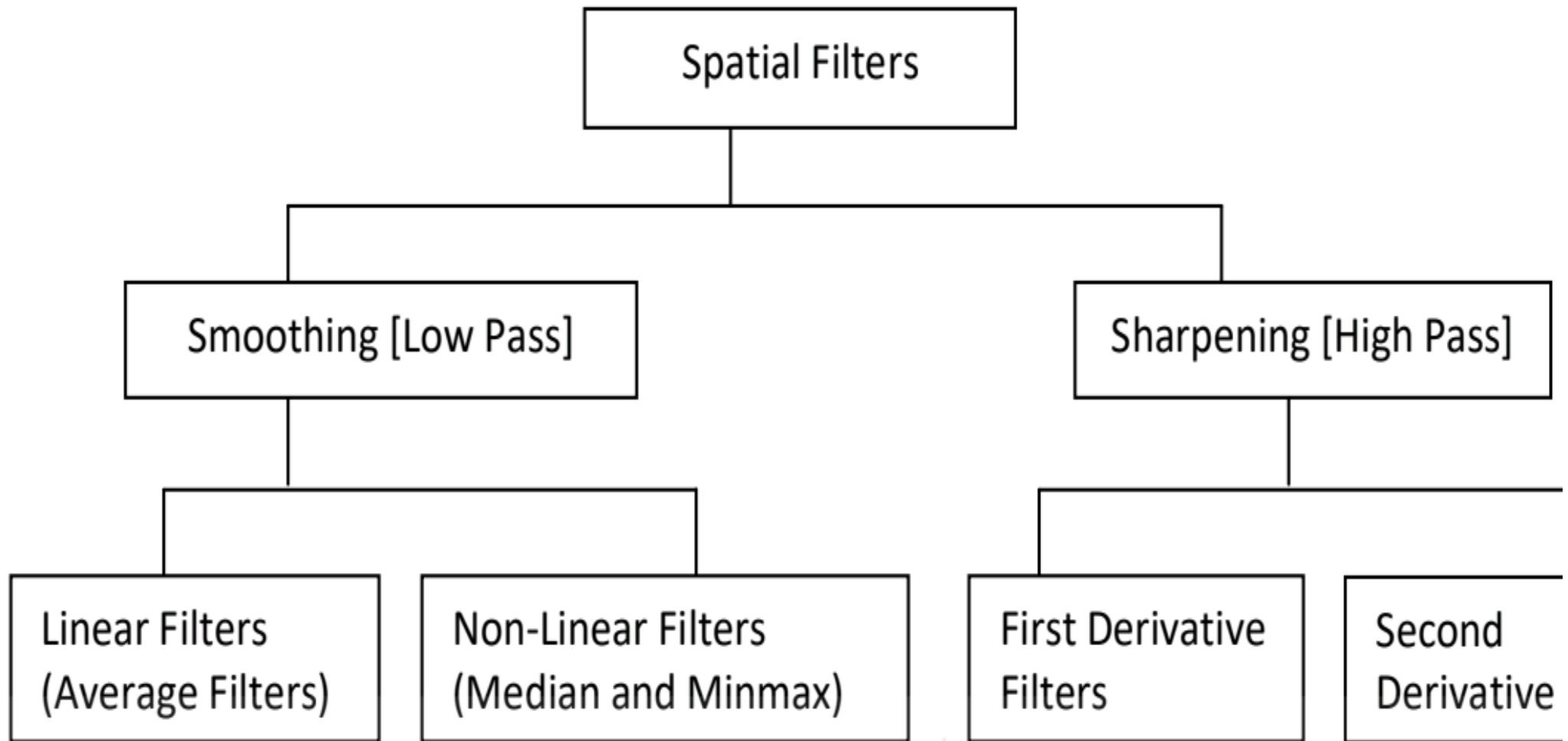
(a) Original Image



(b) Image after filtering

- By this process, we create a new image where grey level values of the pixels are calculated from the values under the mask. The values under the mask are modified by a function called 'filter'.
- If this filter function is a linear function of all grey level values in the mask, then filter is called a '**linear filter**', else it is called '**non linear filter**'.

Types of Spatial Filters



- **Smoothing:** This type of filter is used to blur (smooth) the image, therefore called smoothing filter also. These filters are also used for noise reduction. Noise reduction can be done by blurring with a linear filter, (in which operation performed on the image pixels is linear) or a non-linear filter. ii) **Sharpening:** This filter is used to highlight transitions in intensity. These are based on first and second derivatives.

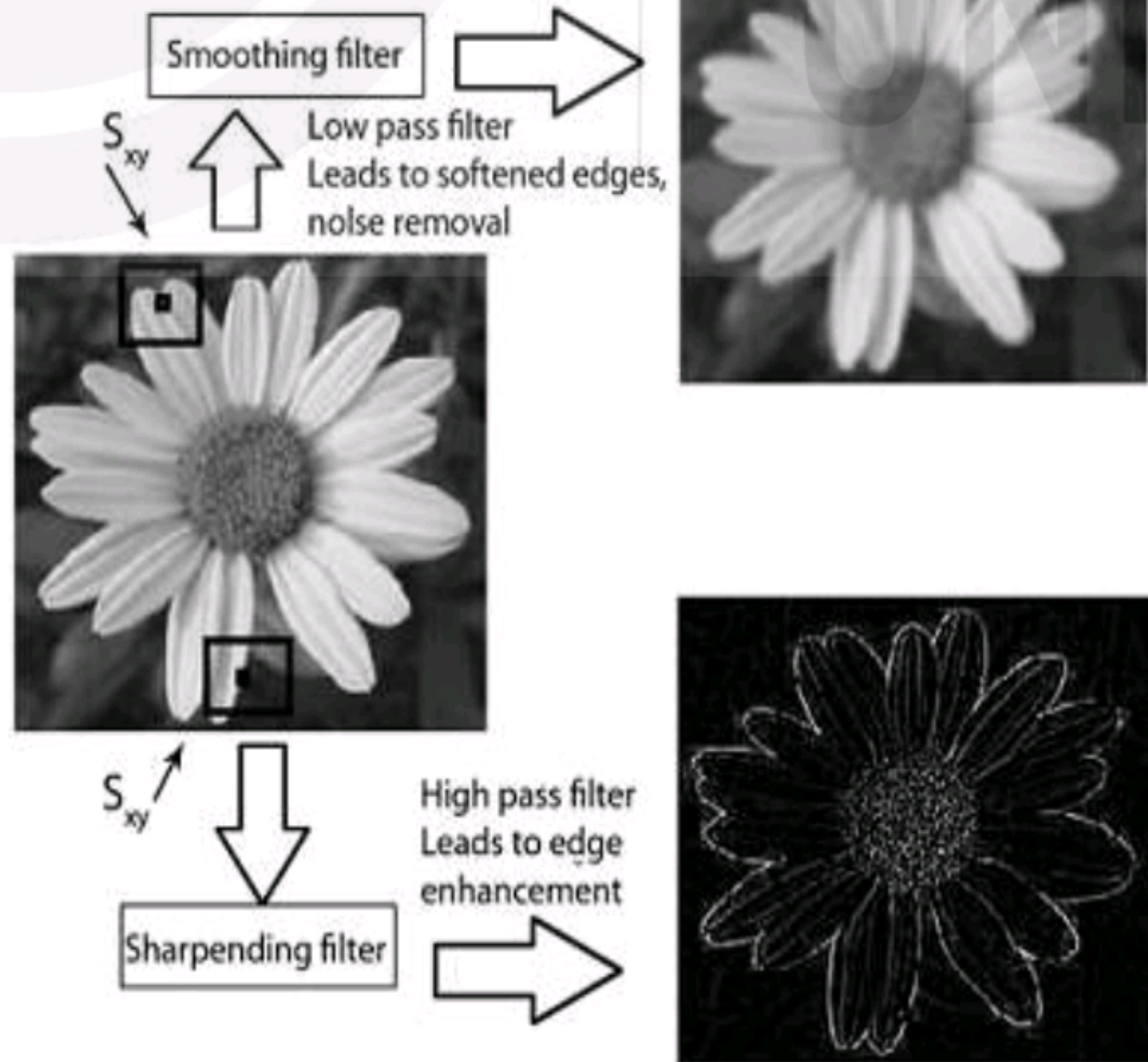


Fig. 3: Smoothing and Sharpening Filters

4.3.1 Linear Filters

Linear filtering is a spatial domain process where a filter (mask/ kernel/ template) with some integer coefficient values is applied to input image to generate the filtered/ output image. Generally, filter size is either 3×3 or 5×5 , 7×7 or 21×21 (odd sizes) and filter is centered at a coordinate (x, y) called '**Hot Spot**' as shown in Fig. 4.

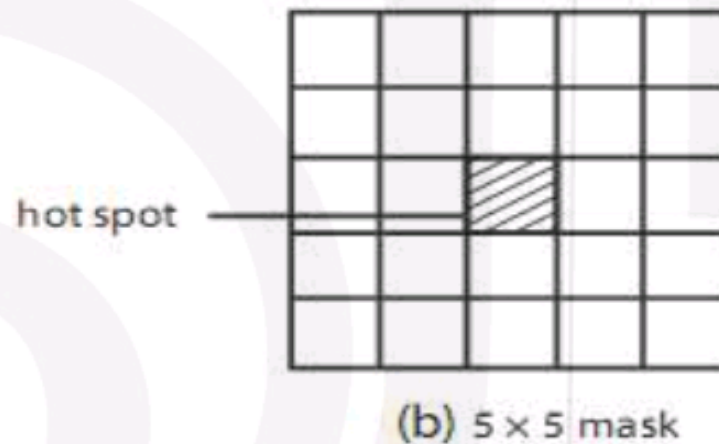


Fig. 4

Linear filtering of an image $f(x, y)$ of size $M \times N$ and filter mask of size $m \times m$ is given by

$$g(x, y) = \sum_{(i,j) \in R_n} f(x+i, y+j) w(i, j) \text{ where } 0 \leq x \leq M-1 \text{ and } 0 \leq y \leq N-1. \quad (1)$$

The right hand side of Eqn. (1) denotes the set of coordinates covered by filter. For a 3×3 filter, with coefficients $w(i, j)$ the output image at coordinate (x, y) is calculated as

$$g(x, y) = \sum_{i=-1}^1 \sum_{j=-1}^1 f(x+i, y+j) w(i, j) \quad (2)$$

linear filtering:

- We carry out the following steps for linear filtering:
- **Step 1:** Position the mask over the current pixel such that hotspot $w(0,0)$ coincides with current pixel.
- **Step 2:** Form all products of filter elements with the corresponding elements in the neighbourhood.

- Step 3: Add up all the products and store it at current position in the output image.
- Step 4: Divide it by the scaling constant (sum of all coefficients of the mask).
- This must be repeated for every pixel in the image.

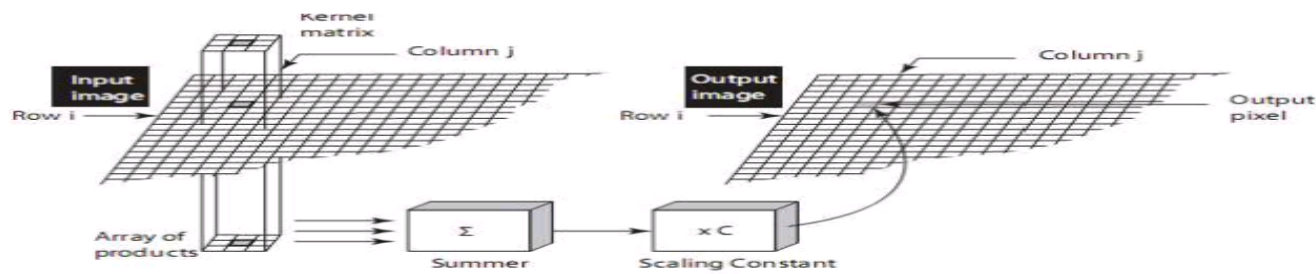


Fig 5: Linear filtering

Let us apply these steps in the following example.

Example 1: Apply given 3×3 mask w on the following image $f(x, y)$.

$$f(x, y) = \begin{bmatrix} 5 & 1 & 2 & 6 & 7 \\ 4 & 4 & 7 & 5 & 8 \\ 2 & 6 & 20 & 6 & 7 \\ 3 & 1 & 2 & 4 & 5 \\ 10 & 2 & 1 & 2 & 3 \end{bmatrix}, w(i, j) = \frac{1}{9} * \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution: First we consider top left 3×3 neighborhood in image $f(x, y)$ as shown in Fig. 6 (a) with hot spot coinciding with grey level 4 in $f(x, y)$ and calculate output for that pixel position by using equation

$$\begin{aligned} \text{Output} &= \frac{1}{9} [5 \times 1 + 1 \times 1 + 2 \times 1 + 4 \times 1 + 4 \times 1 + 7 \times 1 + 2 \times 1 + 6 \times 1 + 20 \times 1] \\ &= \frac{49}{9} \approx 5 \end{aligned}$$

5	1	2	6	7
4	4	7	5	8
2	6	20	6	7
3	1	2	4	5
10	2	1	2	3

(a)

5	1	2	6	7
4	4	7	5	8
2	6	20	6	7
3	1	2	4	5
10	2	1	2	3

(b)

Fig. 6

Shift the mask one pixel towards right such that grey level 7 coincides with hot spot (Fig. 6 (b)). Calculate output for that pixel position and replace in output image. Shift it right again and locate the mask on pixel with grey value 5, and repeat the process.

$$\text{Output} = \frac{1}{9} [1 \times 1 + 2 \times 1 + 6 \times 1 + 4 \times 1 + 7 \times 1 + 5 \times 1 + 6 \times 1 + 20 \times 1 + 6 \times 1]$$

$$= \frac{57}{9} \approx 6$$

Now go back to the first position of the mask and shift it down so that it is now centered on pixel with grey value 6 (just below 4), and repeat the process.

$$\begin{aligned} \text{Output} &= \frac{1}{9}[2 \times 1 + 6 \times 1 + 7 \times 1 + 7 \times 1 + 5 \times 1 + 8 \times 1 + 20 \times 1 + 6 \times 1 + 7 \times 1] \\ &= \frac{68}{9} \approx 8 \end{aligned}$$

Continue doing it over all the possible pixels in the given image.

$$\text{Output} = \frac{1}{9}[4 + 4 + 7 + 2 + 6 + 20 + 3 + 1 + 2] = \frac{49}{9} \approx 5$$

$$\text{Output} = \frac{1}{9}[4 + 7 + 5 + 6 + 20 + 6 + 1 + 2 + 4] = \frac{55}{9} \approx 6$$

$$\text{Output} = \frac{1}{9}[7 + 5 + 8 + 20 + 6 + 7 + 2 + 4 + 5] = \frac{64}{9} \approx 7$$

$$\text{Output} = \frac{1}{9}[2 + 6 + 20 + 3 + 1 + 2 + 10 + 2 + 1] = \frac{38}{9} \approx 4$$

$$\text{Output} = \frac{1}{9}[6 + 20 + 6 + 1 + 2 + 4 + 2 + 1 + 2] = \frac{44}{9} \approx 5$$

$$\text{Output} = \frac{1}{9}[20 + 6 + 7 + 2 + 4 + 5 + 1 + 2 + 3] = \frac{59}{9} \approx 6$$

So, by shifting the mask over the whole image a new image is generated. These values are rounded off to nearest integer values and put in the output image as given below at corresponding pixel locations.

$$g(x, y) = \begin{array}{|c|c|c|c|c|} \hline * & * & * & * & * \\ \hline * & 5 & 6 & 8 & * \\ \hline * & 5 & 6 & 7 & * \\ \hline * & 4 & 5 & 6 & * \\ \hline * & * & * & * & * \\ \hline \end{array}$$

The filtering effect is visible in the new image where the noisy values 10, 20 in original image have been replaced by smoothed values.

- The aim of image smoothing is to diminish the effect of camera noise, spurious pixel values, missing pixel values etc.
- This is done by a neighborhood averaging filter.
- Each pixel in the smoothed image $g(x, y)$ is obtained from the average pixel value in the neighborhood of (x, y) in input image. Such a mask is also known as a Mean filter.

- Smoothing filter is used for the following purposes:
- i) It reduces 'sharp' transitions in grey levels.
- ii) It reduces noise.
- iii) It blurs edges. This is a side effect.
- iv) It helps in smoothing false colours.
- v) It reduces 'irrelevant' details in an image.

- **‘Blurring’** (a side effect of smoothing filter) is used in preprocessing steps for following applications:
- i) Removal of small details from an image prior to object extraction. Intensity of smaller objects blends with the background and larger object become ‘blob like’ and easy to detect.
- ii) Bridging small gaps in lines or curves of the boundaries of objects and text.
- Small gaps in boundary can lead to enormous results in object extraction. Smoothing fills up all smaller gaps and helps in producing correct result.

Nonlinear Filters

- Smoothing linear filters have an important disadvantage, image structures such as individual points, edges and lines are blurred.
- Overall quality of the image reduces in some applications. These side effects are not tolerated which limits the usage of linear filters.

Median Filters

- Median filters are edge preserving smoothing filters, where the level is set to the median of pixel values in the neighborhood of that pixel.
- It is impossible to design a filter that removes only noise and retains all the important image structures intact, because no filter can discriminate which image content
- Median filter replaces every image pixel by median of the pixels in the corresponding filter region S_{xy} .

Advantages of Median Filters

- It has excellent noise reduction capability.
- It preserves edges.
- It introduces less blurring than smoothing linear filters.
- It is very effective in the presence of impulse noise.

Disadvantages

- It performs poorly if number of noise pixels in S
- is greater than half the number of pixels in the window $w \times y$
- It performs poorly in the presence of gaussian noise.

Steps to Implement Median Filter as shown in Fig. 10.

Step 1: Sort the values of the pixel in question and its neighbours.

Step 2: Determine their median value.

Step 3: Assign median value to that pixel.

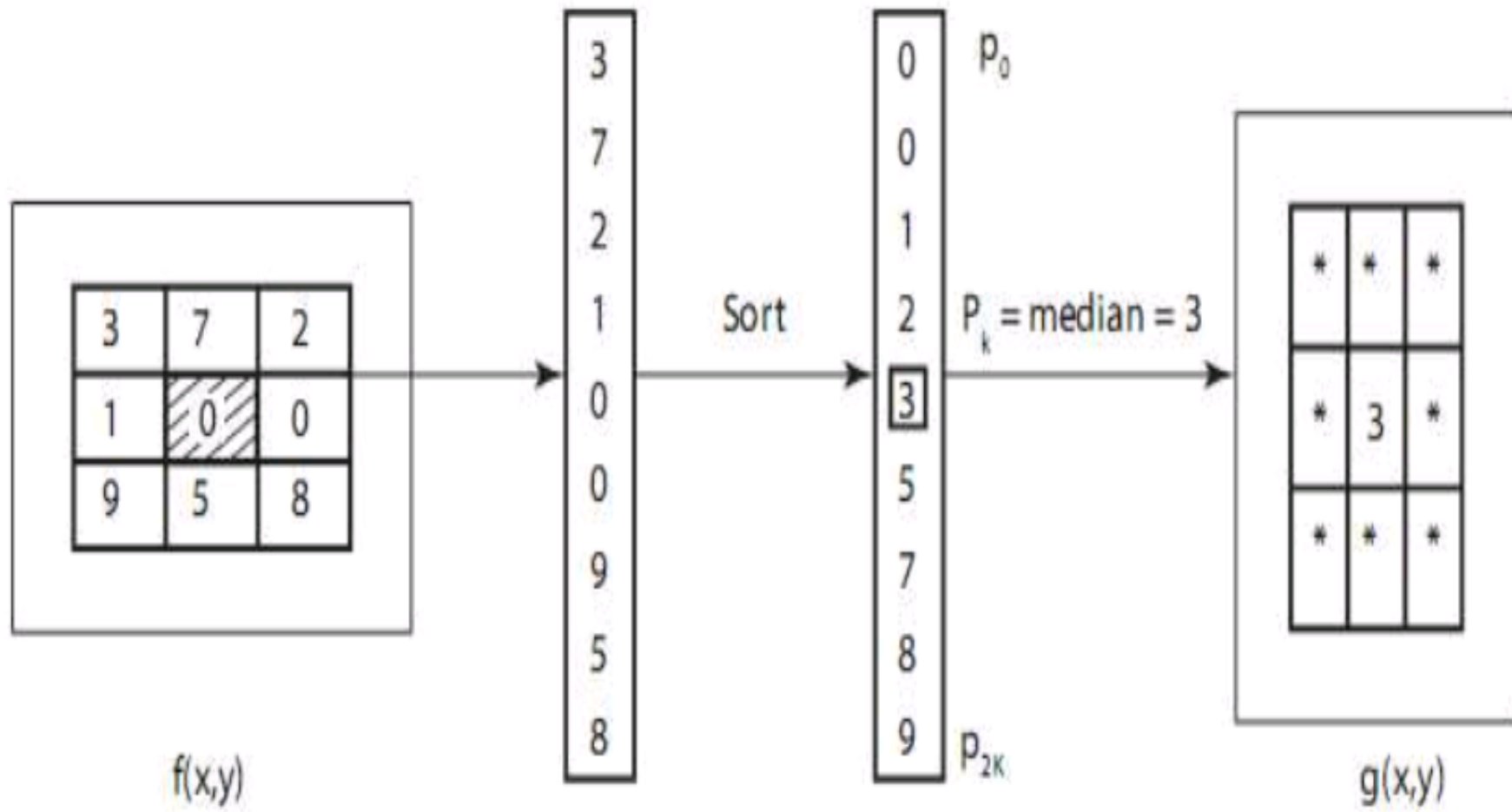


Fig 10: Median Filtering

Example 3: Compute the median value of the marked pixels show below using a 3×3 mask

$$\begin{bmatrix} 18 & 22 & 33 & 25 & 32 & 24 \\ 34 & 128 & 24 & 172 & 26 & 33 \\ 22 & 19 & 32 & 31 & 28 & 26 \end{bmatrix}$$

Solution: i) Median (18, 22, 33, 34, 128, 24, 22, 19, 32)
= Median (18, 19, 22, 22, 24, 32, 33, 34, 128) = 24

ii) Median (22, 33, 25, 128, 24, 172, 19, 32, 31)
= median(19, 22, 24, 25, 31, 32, 33, 128, 172) = 31

iii) Median (33, 25, 32, 34, 172, 26, 32, 31, 28)

$$= \text{median}(24, 25, 26, 28, 31, 32, 32, 33, 172) = 31$$

$$\text{iv) Median}(25, 32, 24, 172, 26, 23, 31, 28, 26)$$

$$= \text{median}(23, 24, 25, 26, 26, 28, 31, 32, 172) = 26$$

Minimum and Maximum Filter

Max and min filters are defined as:

$$g(x, y) = \min \{f(x + i, y + j) \mid (i, j) \in S_{xy}\} \quad (6)$$

$$g(x, y) = \max \{f(x + i, y + j) \mid (i, j) \in S_{xy}\}, \quad (7)$$

where S_{xy} denotes the filter region, usually a size of 3×3 pixels. **Min filter**

removes salt noise (white dots with large grey level values) because any large grey level within a 3×3 filter region is replaced by one of its surrounding pixels with smallest value. As a side effect, min filter introduces dark structures in the image. The reverse effect is expected from a **max filter**. It removes pepper noise (black dots with small grey level values) because any black dot within 3×3 filter region is replaced by one of its surrounding pixels with the largest value. White dots/bright structures are widened as a side effect and black dots (pepper noise) will disappear. Fig. 12 shows the process of min/ max filter.

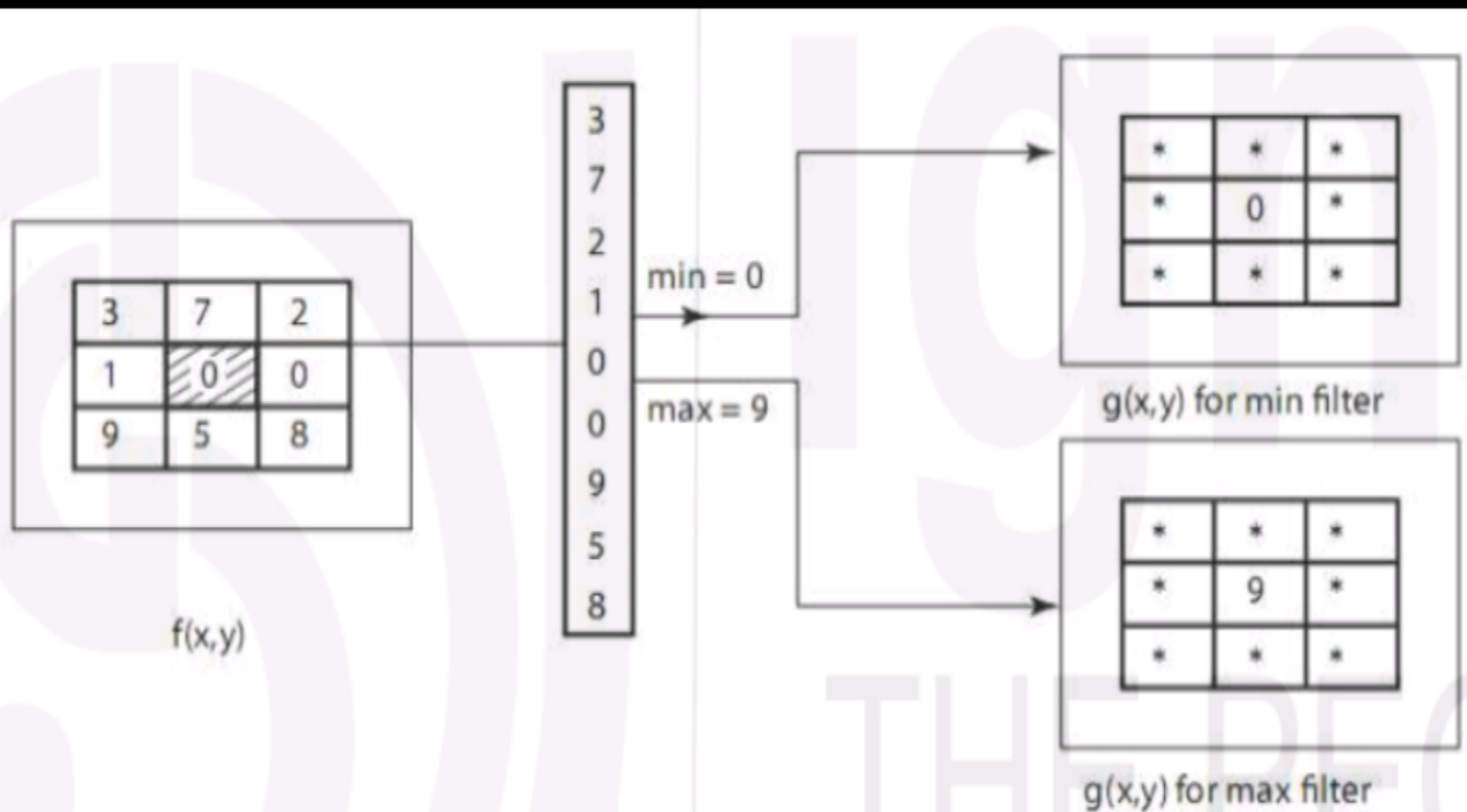


Fig. 11: Min, max filtering

HISTOGRAM PROCESSING

- Histogram of an image represents the number of times a particular grey level has occurred in an image.
- It is a graph in which the x-axis represents gray levels, and y -axis represents number of pixels for each grey level.

Normalized histogram is obtained by dividing the frequency of occurrence of each gray level r_k by the total number of pixels in an image.

$p(r_k) = \frac{n_k}{n}$, where $k = 0, \dots, L-1$, where n = total number of pixels in the image $p(r_k)$ gives the probability of occurrence of grey level r_k . The sum of all components of a normalized histogram is equal to 1.

Example 2: Find normalized histogram of the image

0	0	0	0
0	1	2	3
0	2	4	6

Solution: Fig. 2 image is similar to Example 1. A table is made with r_k, n_k and $p(r_k)$ and normalized histogram is drawn in Fig. 3.

r_k	1	1	2	3	4	5	6
n_k	6	1	2	1	1	0	1
$P(r_k)$	$\frac{6}{12}$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	0	$\frac{1}{12}$

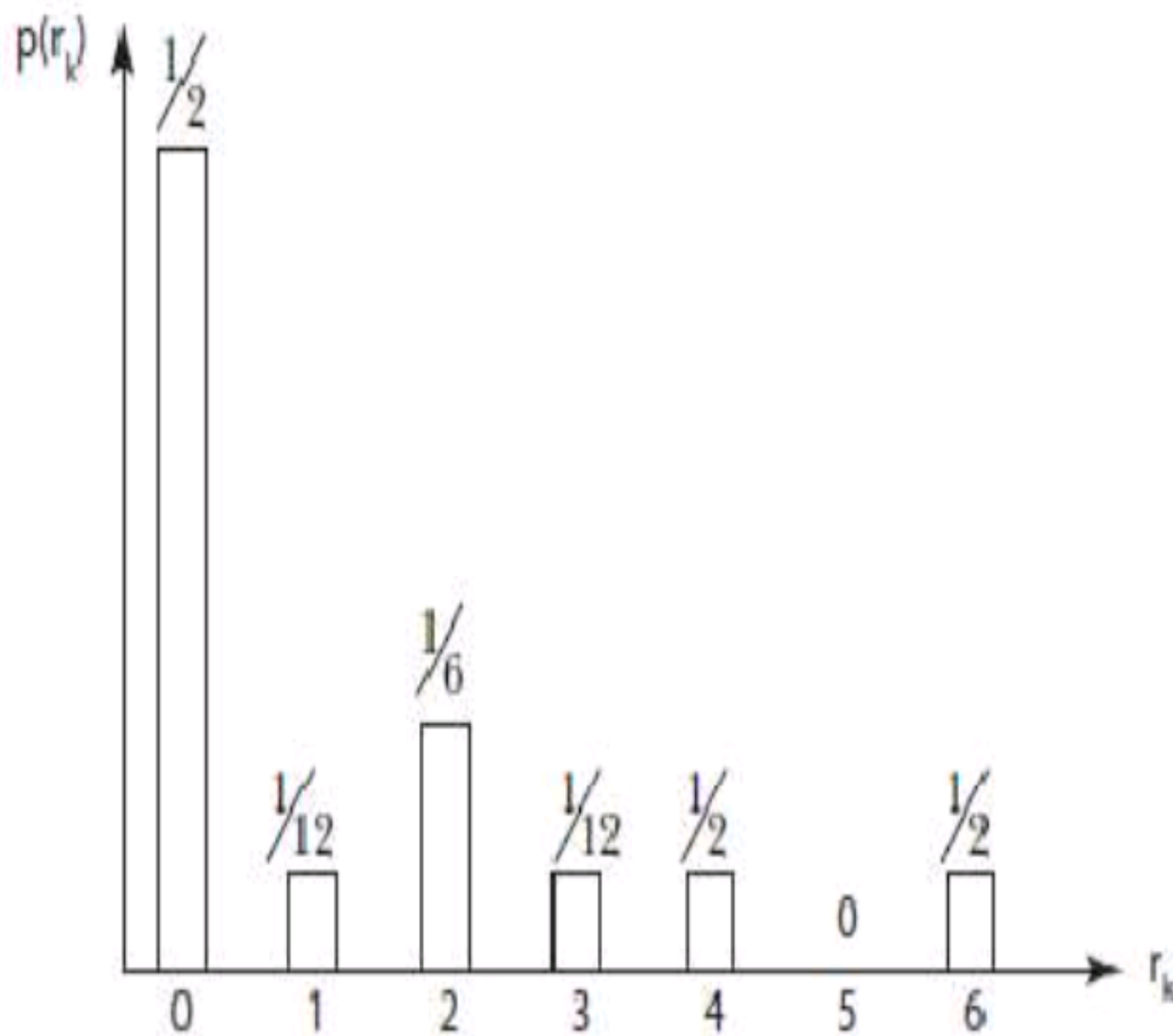


Fig. 3: Normalized Histogram

- Based on histogram, we can categorize images in the four categories, which are defined below.
- i) Under exposed image: Fig. 4 (a) shows an under exposed image, and its histogram is shown in Fig. 4 (b).
- It may be noted that almost all the pixels are concentrated at the lower end of the histogram. The maximum value of grey is less than 50. No details are seen in the image.

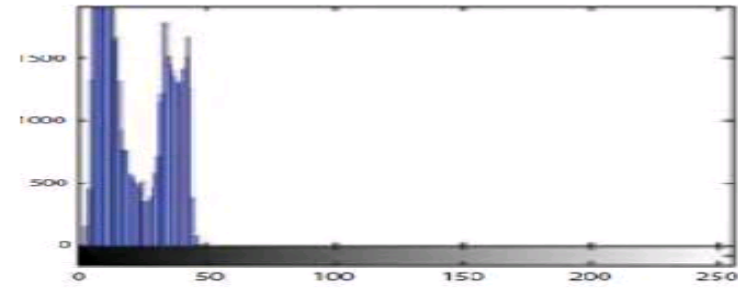
- **Over-exposed image:** Fig. 4 (c) shows an over-exposed image and its histogram is shown in Fig. 4 (d).
- The histogram is prominent towards the higher side of the grey values with the lowest grey level value around 150.
- This image has a washed-out look because of the absence of darker shades of grey.

- **Low contrast image:** Fig. 4 (e) shows a low contrast image and its histogram in Fig. 4 (f).
- The shape of the histogram is narrow and centered towards the middle of the grey scale.
- Very few grey levels participate in the image formation as neither the lower grey levels nor the higher grey levels are present. Thus, details are not visible in the image.

- **High contrast image:** Fig. 4 (g) shows a high contrast image and its histogram in Fig. 4 (h).
- This histogram covers a broad range of grey levels, but distribution of pixels is not uniform.
- The number of some grey levels is very high as compared to the number of other grey levels.



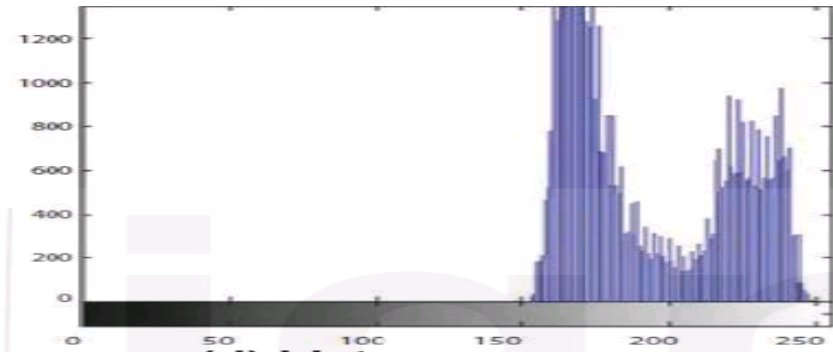
(a) under exposed image



(b) histogram



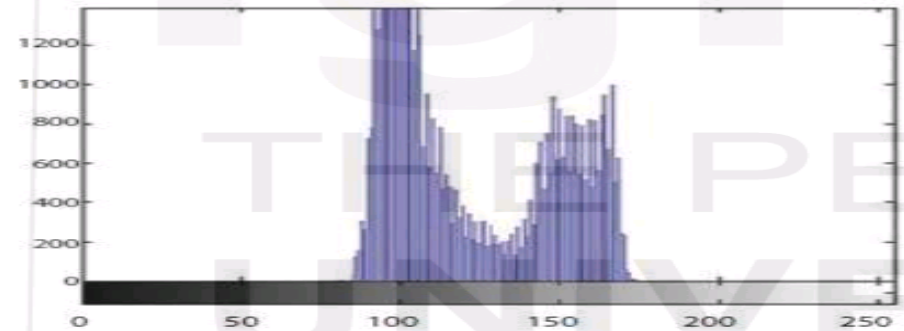
(c) over-exposed image



(d) histogram



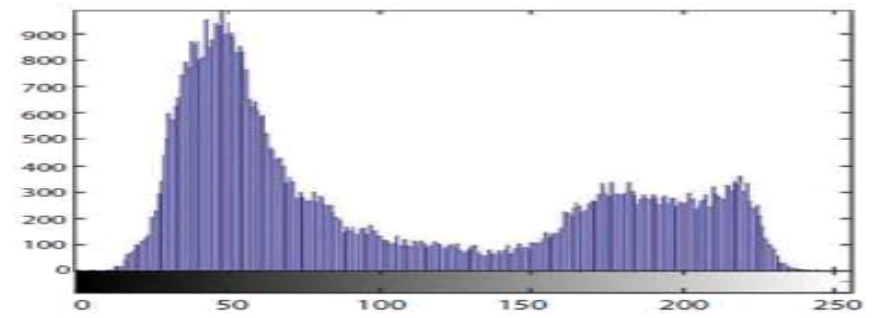
(e) low contrast image



(f) histogram



(g) high contrast image



(h) histogram

Fig. 4

- Histogram indicates if clipping is a problem in the image.
- It is extensively used in thresholding to separate contrasting objects from background.
- It tells us about image contrast.

IMAGE ENHANCEMENT & FILTERING IN FREQUENCY DOMAIN

- In the previous units of this course, we have considered an image in the spatial domain which is the form in which camera captures it.
- In this unit, we shall view the image as a signal and apply the well known filtering techniques used in signal processing.

Types of Frequency Domain Filters

- Frequency domain filters are categorized into three types.
- 1. Smoothing filters
- 2. Sharpening filters
- 3. Homomorphic filters

Frequency Domain Filters

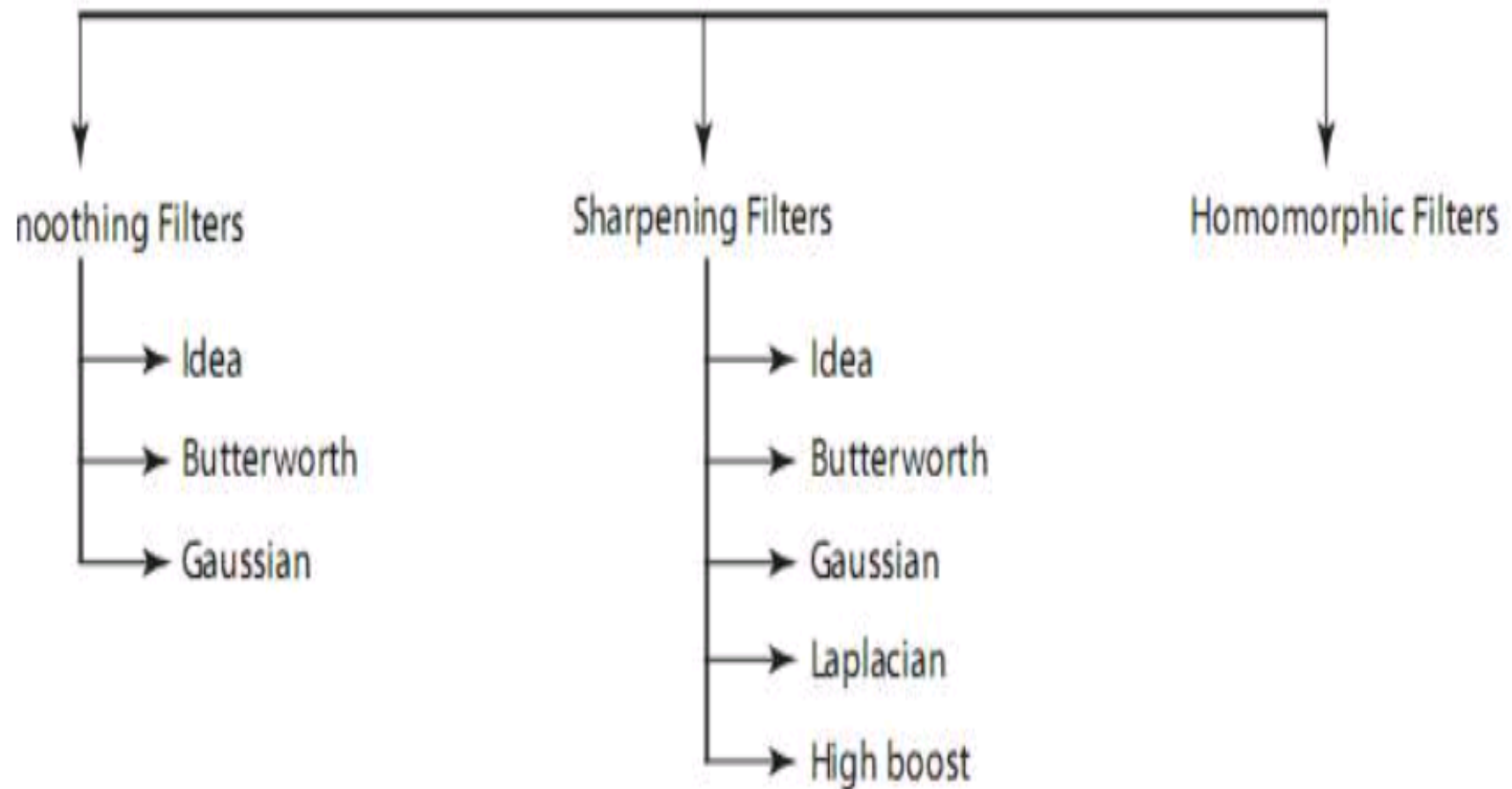


Fig. 5: Types of Frequency Domain Filters.

Smoothing filters

- Smoothing filters are low pass filters (LPF).
- Edges, sharp transitions and noise in the grey levels contribute to high frequency contents in an image. A low pass filter only passes low frequency and blocks the high ones.
- It removes noise but in the process introduces blurring as a side effect in the image..

- The basic model of filtering is

$$G(u, v) = H(u, v) F(u, v)$$

where $F(u, v)$ = Fourier transform of the image to the filtered,

$H(u, v)$ = Transfer function of the filter, and

$G(u, v)$ = Enhanced image where high frequency components have been alternated

- Ideal filters has sharp slope in transition band whereas Gaussian filter has smooth slope in transition.
- Butterworth filter has a parameter called filter order which controls the slope of transition band.
- Higher value of filter order leads to ideal filter.

Ideal Low Pass Filters (ILPF)

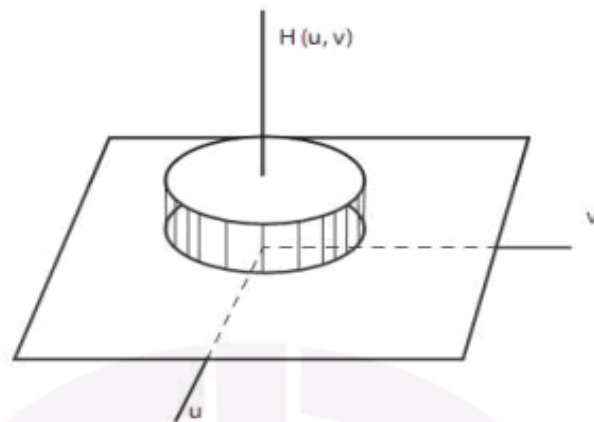
- Ideal Low Pass Filters (ILPF) Low pass filter removes all frequencies above a certain frequency components D_0 .
- Ideal low pass filter is defined by the transfer function

$$H(u, v) = \begin{cases} 1 & D(u, v) \leq D_0 \\ 0 & D(u, v) > D_0 \end{cases}$$

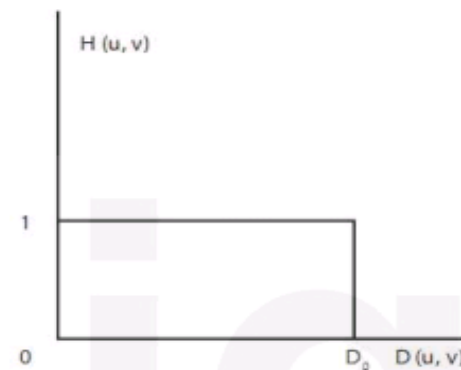
Where $D(u, v) = \left[\left(u - \frac{M}{2} \right)^2 + \left(v - \frac{N}{2} \right)^2 \right]^{1/2}$

$D(u, v)$ is the distance from point (u, v) to the centre $\left(\frac{M}{2}, \frac{N}{2} \right)$. If size of an image is $M \times N$, then the centre is at $\left(\frac{M}{2}, \frac{N}{2} \right)$. Filter transfer function is symmetric about the midpoint.

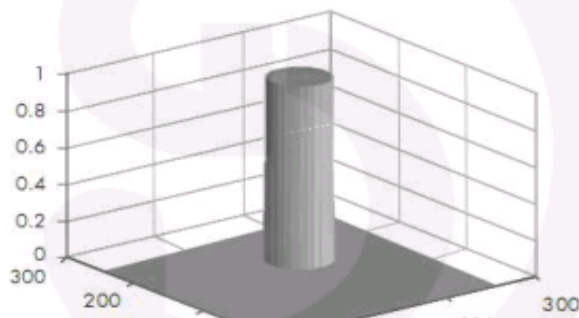
D_0 is non negative quantity specifying the frequency content to be retained. It is also called cut off frequency. Fig. 6(a), Fig. 6(b) is the plot of ILPF and Fig. 6 (c) is perspective plot and Fig. 6(d) is filter displayed as an image.



(a) Transfer Function



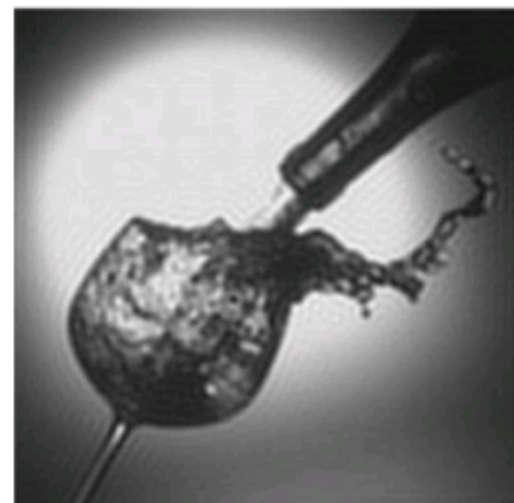
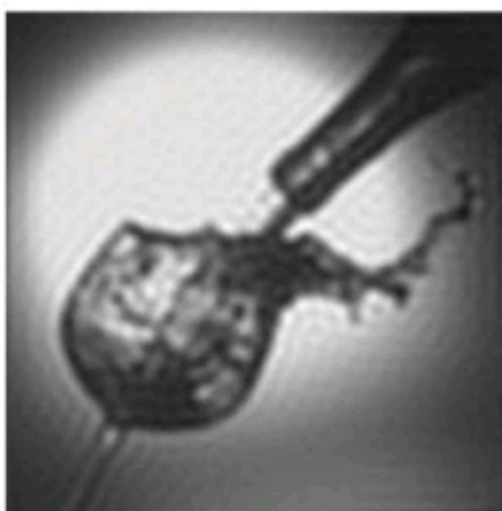
(b) Plot of Ideal LPF



Properties of ILPF

- As it is an ideal filter, it is non-real, non-actual and non-physical.
- But it can be stimulated in computers. Fig 7 and 8(b) to (c) show low pass filtered images with different cut off frequencies.
- As the filter radius increases, less and less power and information is removed which resulted in less blurring.
- A very noticeable effect that can be seen in the output image is ringing.

- Ringing is undesirable and unpleasant lines around the objects present in the image.
- As the cut of frequency D_0 increases, effect of ringing reduces. Ringing is a side effect of ideal lpf .

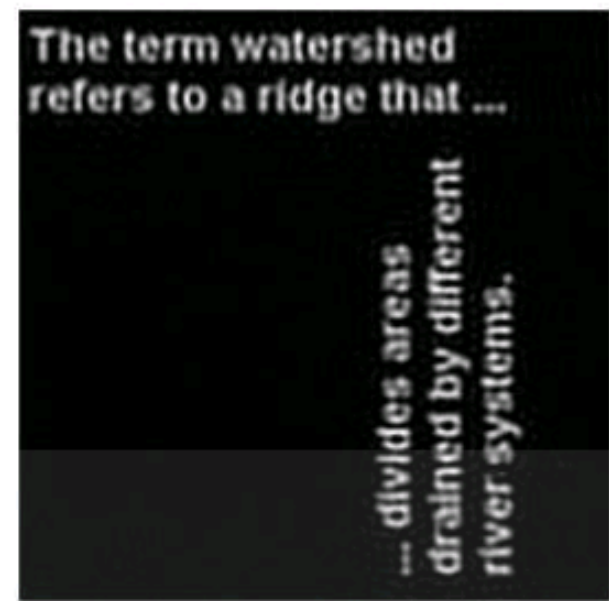


(a) Original image

(b) Output of ILPF with $D_0=30$

(c) Output of ILPF with $D_0=50$

Fig. 7



(a) Original image

(b) Output of ILPF with $D_0=50$

(c) Output of ILPF with $D_0=80$

Fig. 8

IMAGE DEGRADATION

- Overall our objective is to improve image. For that it is important to understand image degradation if we want to remove it. Degradations are of three types
a) Noise b) Blur c) Artifacts
- **Noise** is a disturbance that causes fluctuations in pixel values. Pixel values show random variations and can cause very disturbing effects on the image. Thus suitable strategies should be designed to model and remove/ reduce noise.

- **Blur** is a degradation that makes image less clear. This makes image analysis and interpretation very difficult. Motion blur is a very common cause of blurring where blur occurs due to the movement of object or camera.
- **Artifacts** or distortions are extreme intensity or color fluctuations that can make image meaningless. Distortions involve geometric transformations such as translation, rotation or change in scale.

- The processes that are used to remove degradation are mainly **image enhancement and image restoration**. Restoration is the process of inverting a degradation using knowledge about its nature.
- Whereas enhancement is a process that aims to improve 'bad' quality image so that it looks better. Restoration is distinguished from enhancement, as degradation can be considered as an external influence.

- Major source of noise in digital images is during image acquisition. Nonideal image sensors and poor quality of sensing elements contribute to majority of noise.
- Environmental factors such as light conditions, temperature of atmosphere, humidity, other atmospheric disturbances also account for noise in images.

- Transmission of image is also a source of noise. Images are corrupted with noise because of interference in the channel, lightning and other disturbances in wireless network.
- Human interference also plays a part in addition of noise in images.

Properties of Noise

- Spatial and frequency characteristics of noise are as follows:
- 1) Noise is assumed to be 'white noise' (it could contain all possible frequency components), as such, fourier spectrum of noise is constant.
- 2) Noise is assumed to be independent in spatial domain. Noise is 'uncorrelated' with the image, that is, there is no correlation between pixel value of image and value of noise components.

Types of Noise Model

- Noise models are divided into two categories:
- a) Noise which is independent of spatial location: Gaussian, Rayleigh, Gamma, Exponential, Uniform noise are examples of this category.
- b) Noise which is dependent on spatial location: Periodic noise is example of this type of noise.

Gaussian Noise

- Gaussian noise model is most frequently used in practice. The PDF of a Gaussian random variable 'z' is given by

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}, \quad (2)$$

where z = intensity/grey level value

μ = mean (average) value of z

σ = standard deviation

Plot of $p(z)$ with respect to z is shown in Fig. 26. 70% of its values are in the range $[(\mu - \sigma), (\mu + \sigma)]$ while 95% of the values are in the range $[(\mu - 2\sigma), (\mu + 2\sigma)]$. DFT of **Gaussian (normal) noise is another Gaussian process**. This property of Gaussian noise makes it most often used noise model. Some examples where Gaussian model is the most appropriate model are electronic circuit noise, sensor noise due to low illumination or high temperature, poor illumination.

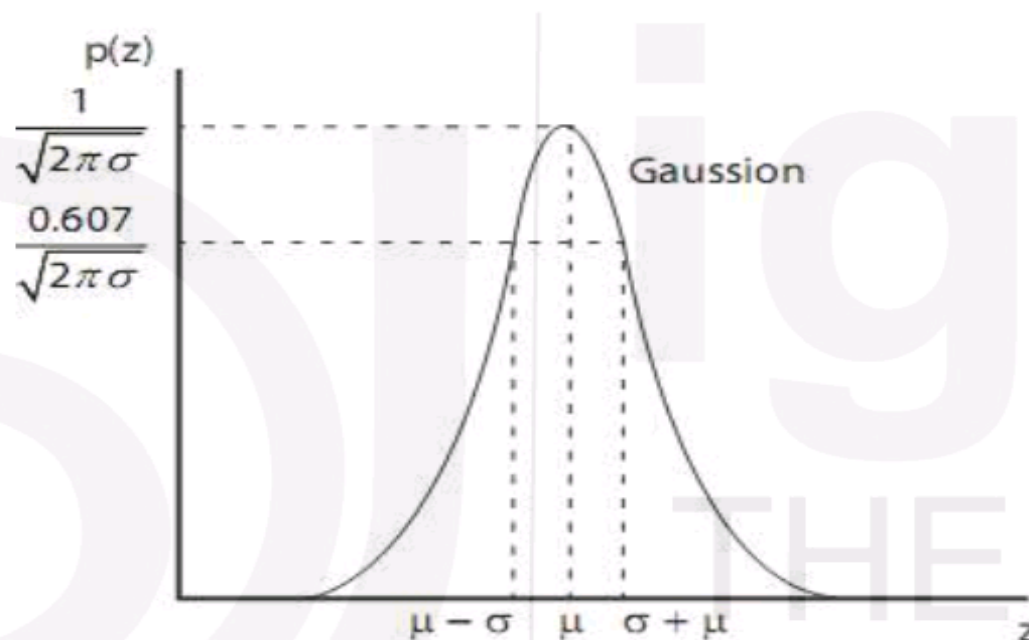


Fig. 26: PDF of Gaussian Noise Model

- Rayleigh Noise Radar range and velocity images typically contain noise that can be modeled by the Rayleigh distribution. Rayleigh distribution is defined by

$$p(z) = \begin{cases} \frac{2}{b}(z-a)e^{-\frac{(z-a)^2}{b}} & z \geq a \\ 0 & z < a \end{cases}, \quad (3)$$

Mean density is given as $\mu = a + \sqrt{\pi b/4}$ and variance is given by

$$\sigma^2 = \frac{b(4-\pi)}{4}.$$

- Plot of PDF is shown in Fig . As it is clear that the curve doesn't start from origin and is not symmetrical with respect to the centre of the curve. This is mainly used in range imaging.

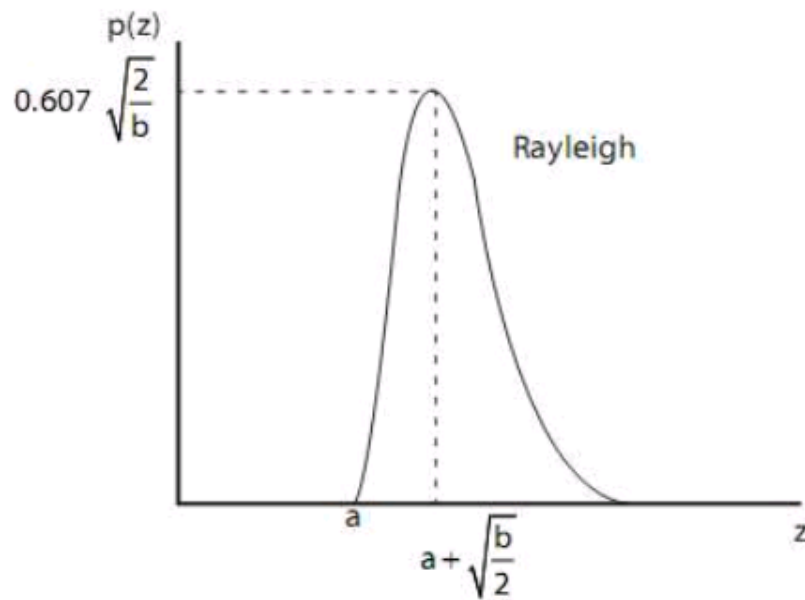


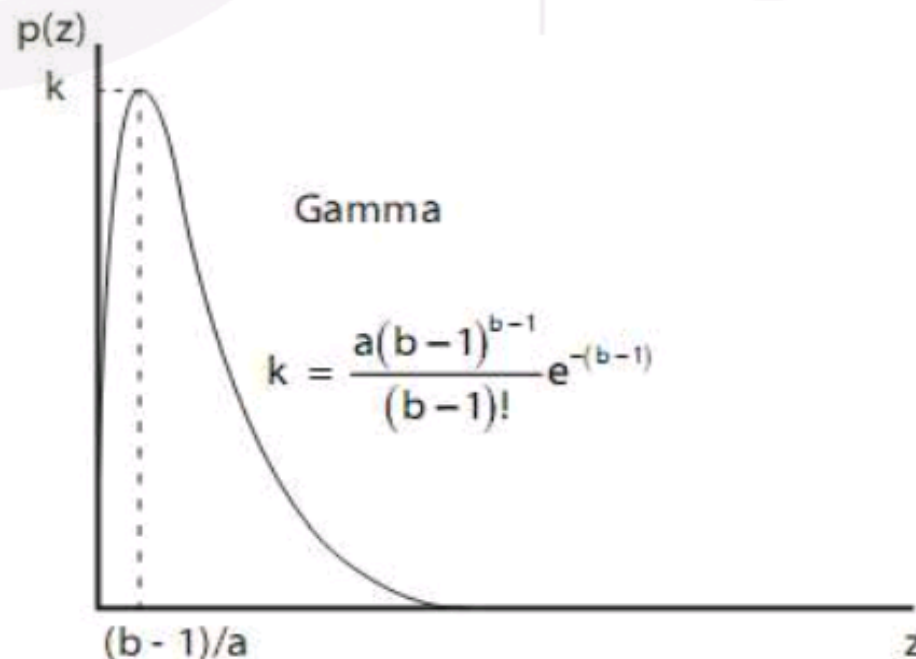
Fig. 27: PDF of Rayleigh Noise

Erlang noise is given by

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}, \quad (4)$$

Where a and b are positive integers, mean density is given by $\mu = \frac{b}{a}$
and variance is $\sigma^2 = \frac{b}{a^2}$.

When the denominator is a gamma function, the PDF describes the gamma distribution. Plot is shown in Fig. 28.



Uniform Noise

- Uniform noise is specified as
 $P(z) = 1/(b-a)$ if $a \leq z \leq b$ and 0 otherwise
Then mean and variance of uniform noise is

$$\mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}$$



Fig. 29: PDF of Uniform Noise.

Fig. 29 shows the plot of PDF of uniform noise. Uniform noise is least used in practice.

Impulse (Salt and Pepper) Noise

- Impulse (Salt and Pepper) Noise Impulse (salt and pepper) noise is specified as

Impulse (salt and pepper) noise is specified as

$$p(z) = \begin{cases} P_a & z = a \\ P_b & z = b \end{cases},$$

(6)

- Fig. 30 shows the plot of PDF of impulse noise. If $b = a$, intensity (grey level) 'b' will appear as a light dot on the image and 'a' appears as a dark dot. This is a 'bipolar' noise, If $P \geq 0$ or $P \leq 0$ unipolar noise
Generally, a and b values are saturated (very high or very low value), resulting in positive impulses being white (salt) and negative impulses being black (pepper).

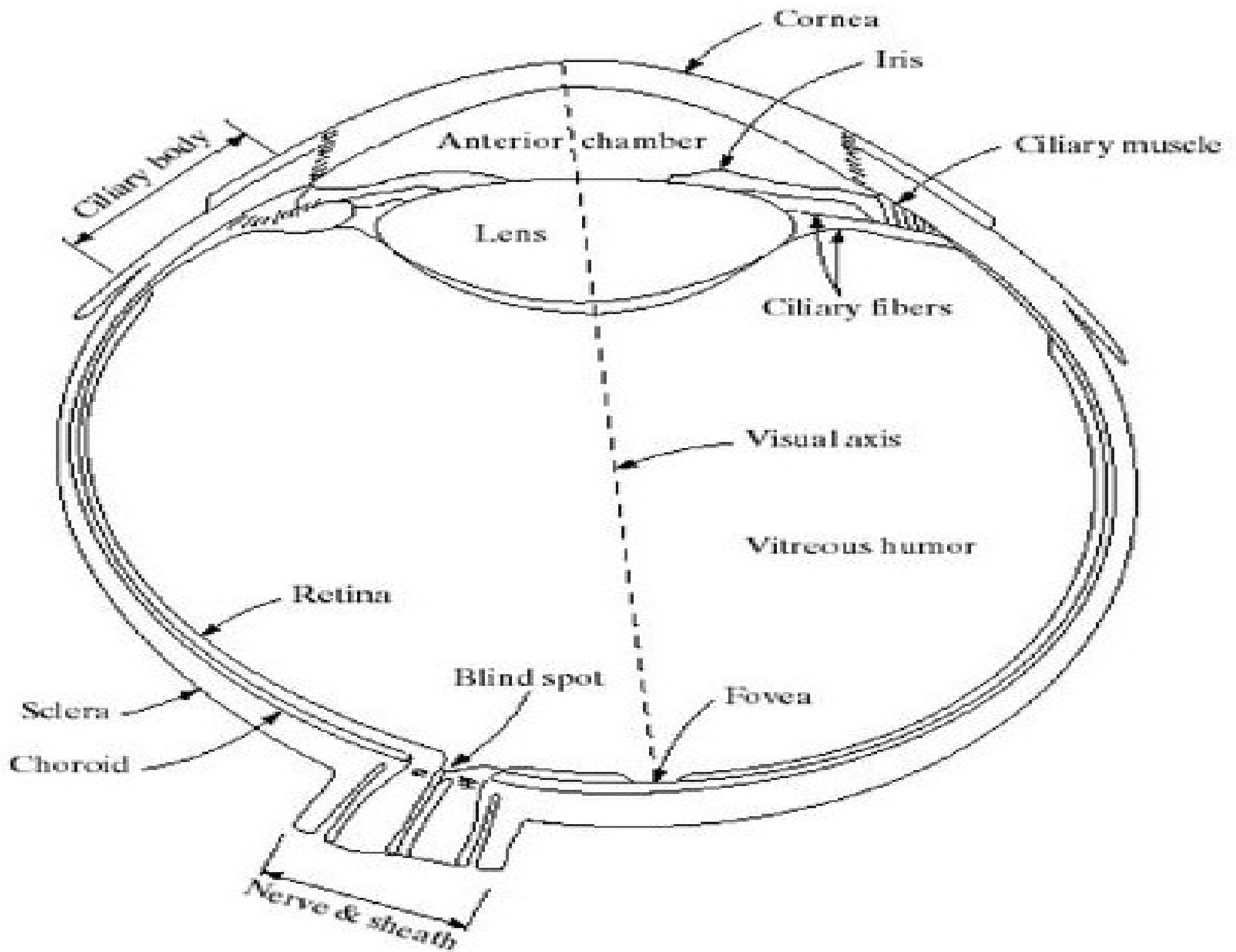
- If $P(0, a) \neq 0$ and $P(b, 0)$ exists, this is called 'pepper noise' as only black dots are visible as noise. If $P(0, b) \neq 0$ only $P(a, 0)$ exists, this is called 'salt noise' as only white dots are visible on the image as noise. Impulse noise occurs when quick transitions happen, such as faulty switching takes place. Noise parameters are generally estimated based on histogram of small flat area of noisy image.

- The salt & pepper noise is generally caused by malfunctioning of camera's sensor cells, by memory cell failure or by synchronization errors in the image digitizing or transmission.

HUMAN VISION SYSTEM

The human eye is nearly spherical in shape with a 20mm diameter on an average, as shown in Fig. The three main parts of the eye are:

- i) The cornea and sclera outer cover,
- ii) the choroid and
- iii) the retina.



i) As we see in Fig, the sclera is an opaque member that encloses the optic globe all around, except at the anterior end, which is covered by the cornea.

The cornea is a tough, transparent cover of the anterior chamber.

ii) Choroid is the layer under the sclera. The membrane choroid contains a network of blood vessels. These blood vessels form the major source of nutrition for the eyes.

If the choroid is damaged and inflamed, it can restrict blood flow in the eye, resulting in serious damage of the eye cells. The role of the choroid is also to control the amount of light entering the eye as well as reduce the backscatter inside the eye.

The choroid is divided into two parts:

a. The ciliary muscles which relax and tighten to enable the lens to focus by changing its shape.

b. The iris diaphragm, that contracts and expands to control the amount of light that enters the eye. The lens is a transparent, biconvex structure that helps to refract light into the eye such that the image is formed on the retina. The lens is flexible and can change shape to change the focal length of the eye. This ensures that objects at various distances can be focussed upon and their images can be formed on the retina.

The retina is the innermost membrane of the eye. It lines the wall of the complete posterior portion of the eye. The retina can be thought of as the image plane in the eye, since on properly focussing the eye on an object, light from that object is passed through the lens such that the image is formed on the retina.

The retina consists of two types of cells called rods and cones.

The cones are highly sensitive to colour and are around 6-7 million in a human eye. The cones are located on the fovea which is the central portion of the retina.

137
01
36

However, there are 75-150 million rod cells which are completely

distributed all over the retina. The rod cells give the overall

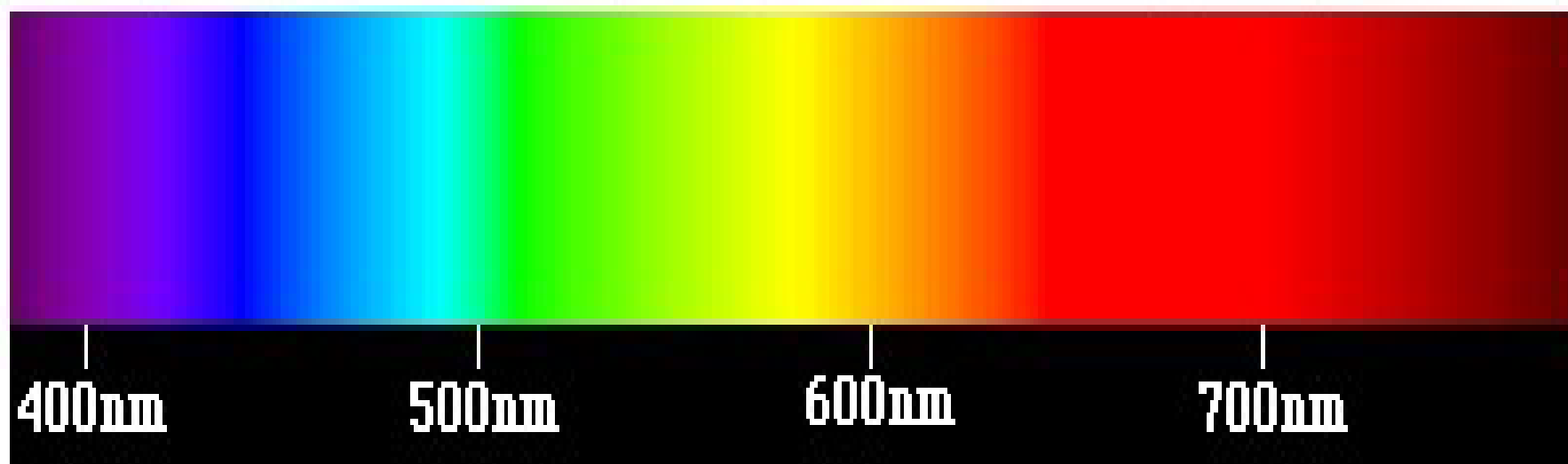
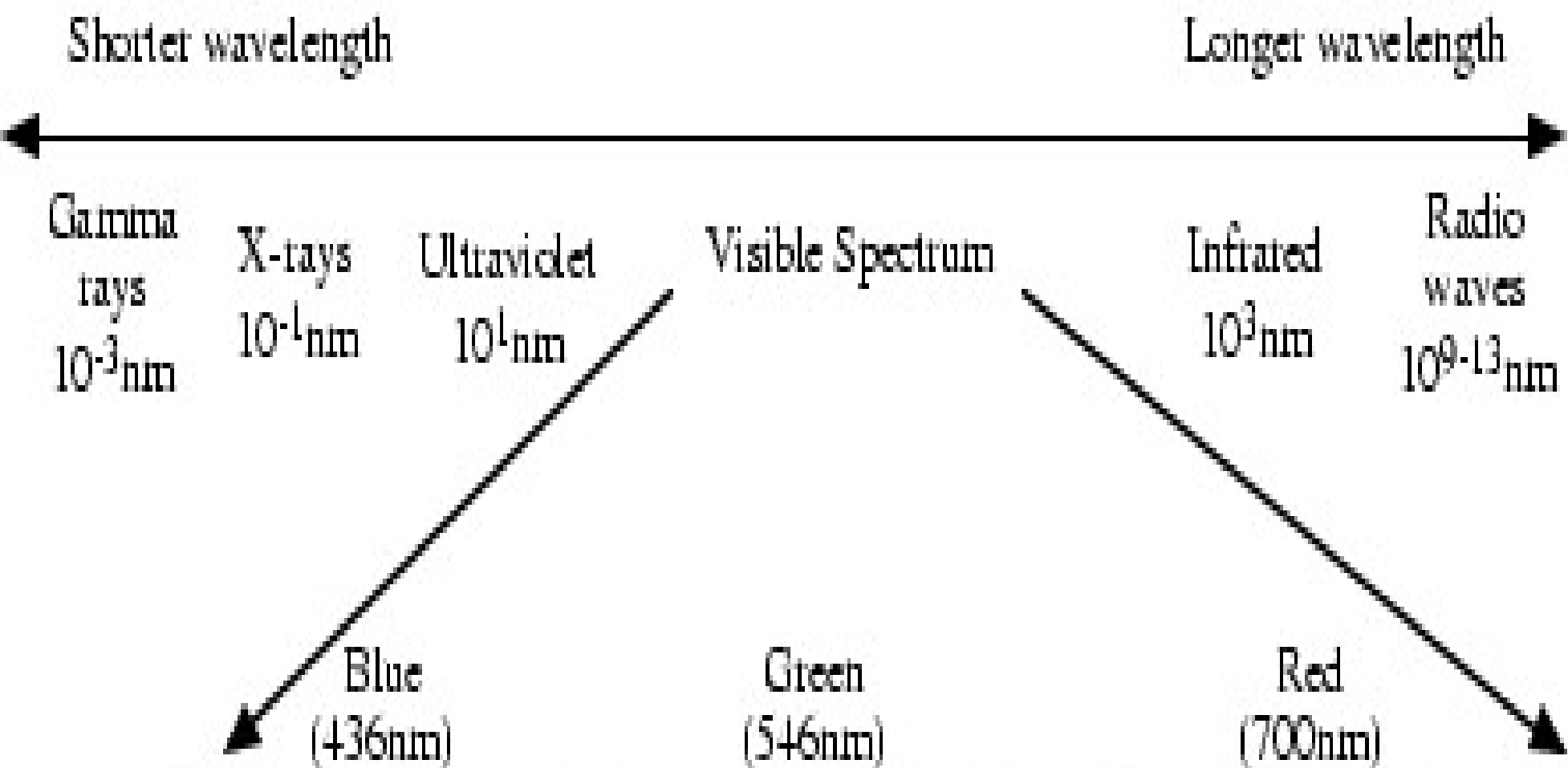
picture of the object in the scene, and reduce the amount of detail. Rods are also responsible for low light vision, also known as SCOTOPIC vision, while cones are responsible for bright light vision, also known as PHOTOPIC vision.

In the human eye, the distance between the retina and lens, that is the focal length, varies between 14 and 17 mm as the refractive power of the lens increases from min to max. For a nearby object, the lens is most strongly refractive. Moreover, the lens of the eye is very flexible and is flattened by controlling

COLOUR FUNDAMENTALS

Every colour is defined using three quantities that are independent of each other, however, taken together they define a particular shade of a colour. These quantities are:

i) Hue: Hue component of a colour is defined by the dominant wavelength. The wavelength range on the electromagnetic spectrum that defines the visible colour spectrum lies between 400nm [nm is nanometre] that represents the violet colour and 700nm that represents the red colour as can be seen in Fig.



- 140
36
- ii) Saturation: The excitation purity of the colour is determined by the quantity known as saturation. It is dependent on the amount of white light that is mixed with hue of that colour. A fully saturated colour implies that no white light is mixed with that hue.
 - iii) Chromaticity: The sum of hue and saturation constitutes the chromaticity of the colour. Therefore, if there is no colour, it is called achromatic light.
 - iv) Intensity: The amount of light actually present defines the intensity.
- Therefore, intensity is a physical quantity. If more light is present, the colour is more intense. Achromatic light has only intensity but no colour. Grayscale images have only intensity

v) **Luminance or Brightness:** The perception of colour is the quantity

known as luminance or brightness. For example, given two colours

of the same intensity, such as blue and green, it is perceived that blue is much darker than green.

vi) **Reflectance:** The ability of an object to reflect light, is the reflectance property of the object. The reflectance property determines the colour of the object, since, we see those colours that

are reflected back and not the ones that are absorbed. For example,

an object that reflects green, absorbs all other colours in the white light spectrum except green.

There are about 6 to 7 million cones in the human eye and they are responsible for recognising colours. Nearly, 65% of the cones recognise

red, 33% are sensitive to green and about 2% to blue.

Red, green and blue are known as the primary colours and nearly all other colours are seen as a combination of these primary colours.

However, there is a difference between the primary colours of light and the primary colours of pigments. The primary colours of light are red, blue and green and they can be added to produce the secondary colours of light that are yellow (red plus green), magenta (red+blue) and cyan (green + blue).

Moreover, the primary colours of pigments are said to be those that

absorbs a primary colour of light and reflects the other two.

Therefore,

in this case, the primary colours are cyan, magenta and yellow.

143
OF
36

For standardisation, in 1931, the Commission Internationale de l'Eclairage (CIE) defined specific wavelengths for the three primary

colours: red = 700nm, blue: 435.8 nm and green = 546.1 nm.

The amounts of red, blue and green required to form a colour are known as the Tristimulus values and are denoted as X,Y and Z.

Given X,Y and Z the tristimulus coefficients which define a colour. If we define x, y and z as the relative values of the primary colours, then

these values can be found by

$$X=X/(X+Y+Z) \quad Y=Y/(X+Y+Z) \quad Z=Z/(X+Y+Z)$$

It is obvious that $x + y + z = 1$.

If we specify colours as a composition as x (red) and y (green). Then.

Example 1: Consider the coordinates of warm white (0.45, 0.4) and the

coordinates of deep blue (0.15, 0.2). Find the percentage of the three colours red (X), green (Y) and blue (Z).

Solution: We first find the trichromatic coefficients x, y and z. At the

point warm white, $x = 0.45$ and $y = 0.4$, therefore

$$\begin{aligned} z &= 1 - (x + y) \\ &= 1 - (0.45 + 0.4) \\ &= 0.15 \end{aligned}$$

Now, we shall find the tristimulus values X, Y and Z.

$$\begin{aligned} X &= X / (X + Y + Z) = 0.45 \\ Y &= Y / (X + Y + Z) = 0.40 \\ Z &= Z / (X + Y + Z) = 0.15 \end{aligned}$$

Here, $X : Y : Z = 0.45 : 0.4 : 0.15$

Therefore the percentage of each colour would be as follows:

- Percentage of red (X) = 45%
- Percentage of green (Y) = 40%
- Percentage of Blue (Z) = 15%

COLOUR MODELS

The principles of absorption of colours of any model by human eye.

i) The human eye has absorption characteristics of colours and recognises them as variables. Thus, the colours red (R), green (G) and blue

(B) are called primary colours of light.

ii) Secondary colours of light are produced by adding primary colours. For example red and blue produces magenta, red and green produces yellow, green and blue produces cyan, etc.

iii) Proportion of primary and secondary colours in appropriate amount produces white light.

The RGB Model

The RGB colour is based on a cartesian coordinate system, where the

colour subspace is a cube with axes representing red, green and blue.

A colour in the RGB model is therefore, specified as a 3-tuple (R, G, B)

where, R, G and B represent the amount of red, green and blue, respectively, present in that colour.

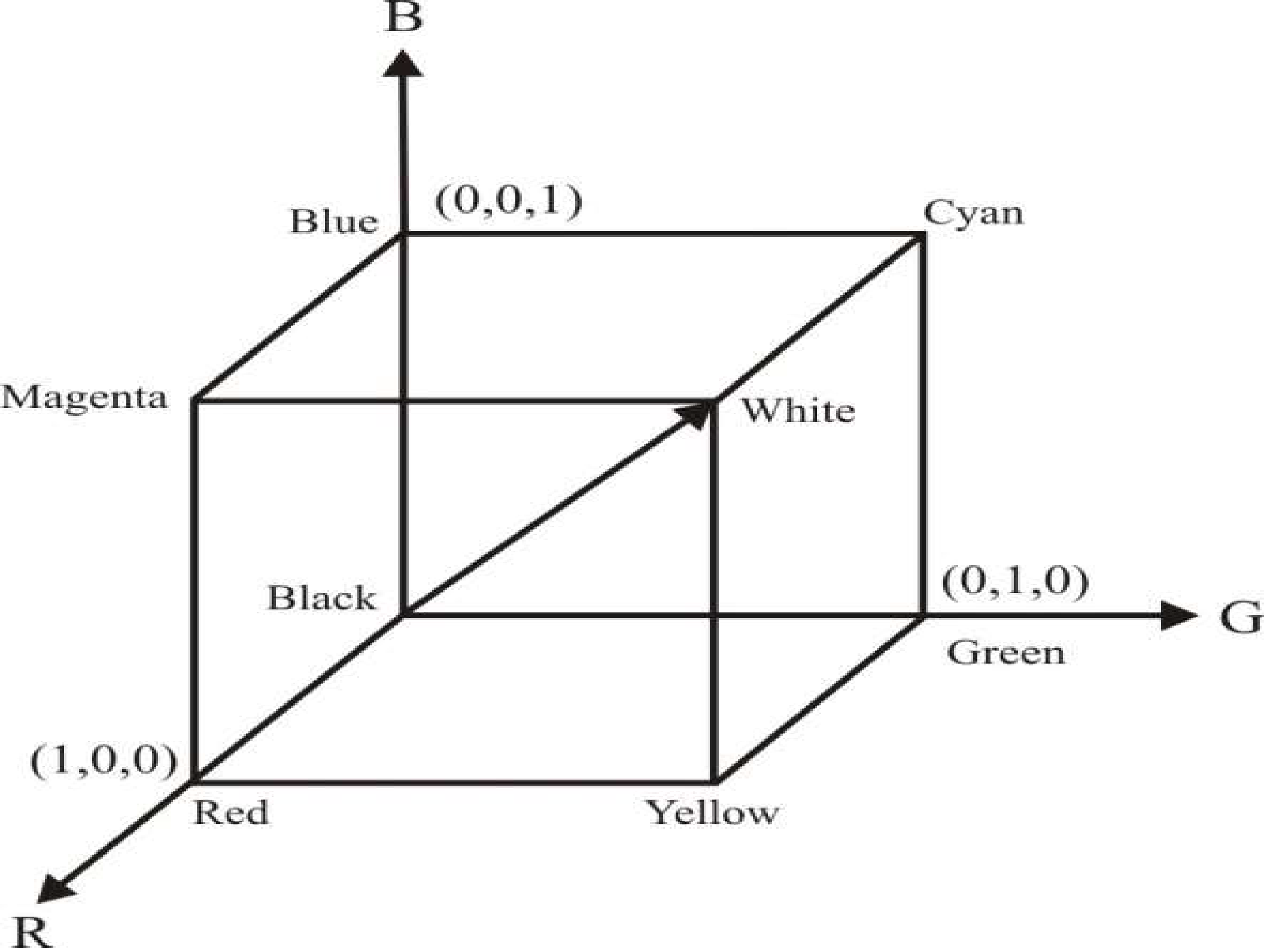
The geometry of the RGB colour model is a cube as shown in Fig. . It is assumed that all colour values

are normalised, that is, it is a unit cube and all values of R, G and B lie

between 0 and 1.

It is clear from Fig. , that it is a model of a cube with the three axes for Red, Green and blue.

The grayscale values lies on the diagonal of the cube, which joins the black (0.0.0) and white (1.1.1)



A colour image in the RGB model consists of three images corresponding to each of the three colours: Red (R), Green (G) and Blue (B) colours. These three images combine to form one composite colour image on a monitor. To convert an RGB image to a grayscale image, the intensity of the gray-pixel is given by the average of R, G and B values. The RGB colour model is mainly used for colour monitors and screens.

Now the question arises how do we find the composite colour in RGB colour model at any point. For this we follow the following steps:

Step 1: Pixel depth is the number of bits used to represent each pixel. If an image in RGB model has 8-bit image in each of its three colours, then each RGB pixel has a depth of 3 image planes $\times$ 8-bit per plane that is 24 bits. This gives rise to 224 colour shades.

Step 2: We fix one of the three colours and let the other two colours to vary. Suppose we fix $R \in [0, 255]$ and let G and B to vary. Then the colour at any point on the plane parallel to GB plane would be (R, G, B) , where, $G, B \in [0, 255]$.

Example 3: In a RGB image, the R and B components are at mid and the G component is at 1, then which colour would be seen by a person?

Solution: At the given point, we have

$$\begin{aligned} R/2 + B/2 + G &= \frac{1}{2}(R + B + G) + G/2 \\ &= \text{midgray} + \frac{1}{2} G = \text{Pure green with some grey component} \end{aligned}$$

The CMY and CMYK Colour Model

Cyan (C), Magenta (M) and Yellow (Y) are the primary colours of pigments and the secondary colours of light. The CMY model is a subtractive model, implying that it subtracts a colour from the white light

and reflects the rest.

For example, when white light is reflected on cyan, it subtracts Red and reflects the rest. While RGB is an additive model,

where something is added to black (0,0,0) to get the desired colour

CMY is a subtractive model. The conversion between CMY and RGB

model is given by the Equation below.

$$\begin{array}{rcl} C & 1 & R \\ M & = & 1 - B \\ Y & 1 & G \end{array}$$

151
of
66

where, the RGB values have been normalised. This also gives a method to convert from RGB to CMY to enable printing hardcopy, since

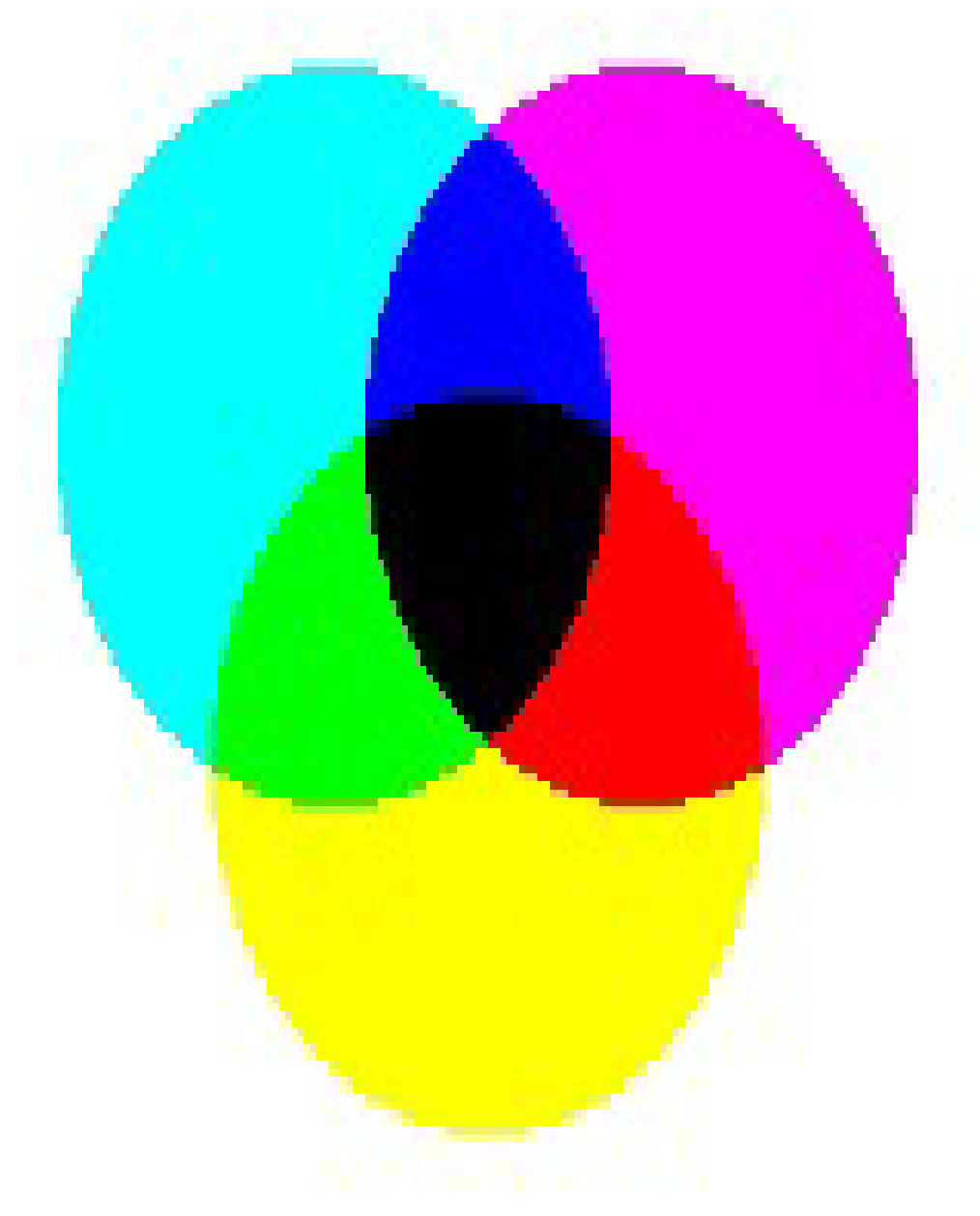
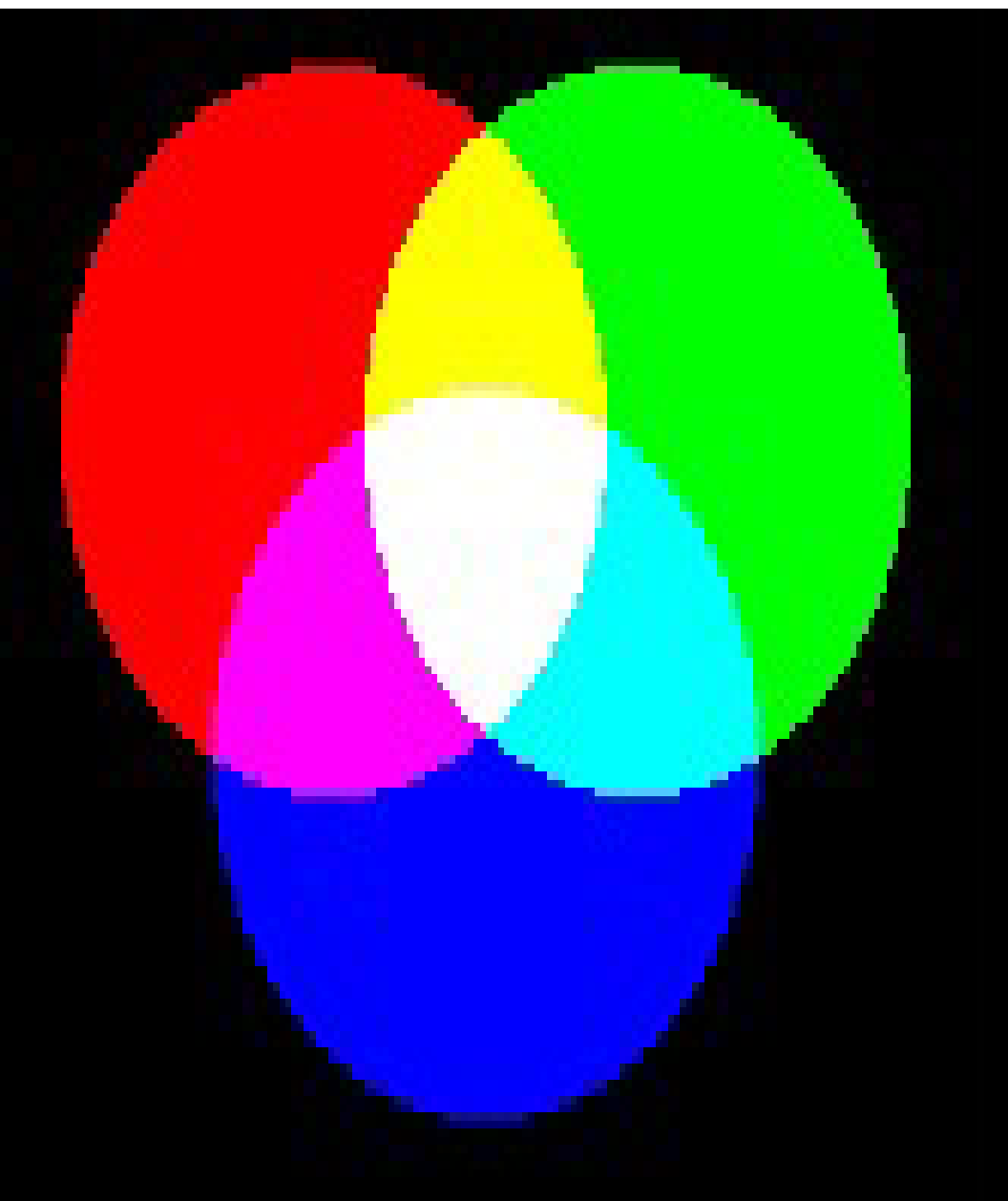


Fig. (a) shows the RGB model, in which the white colour is produced by adding the three primary colours Red, Green and Blue. Fig. (b) shows the CMY model, where black is obtained as the sum of Cyan, Magenta and Yellow. The inverse relation between the RGB and CMY models are also shown by these two images.

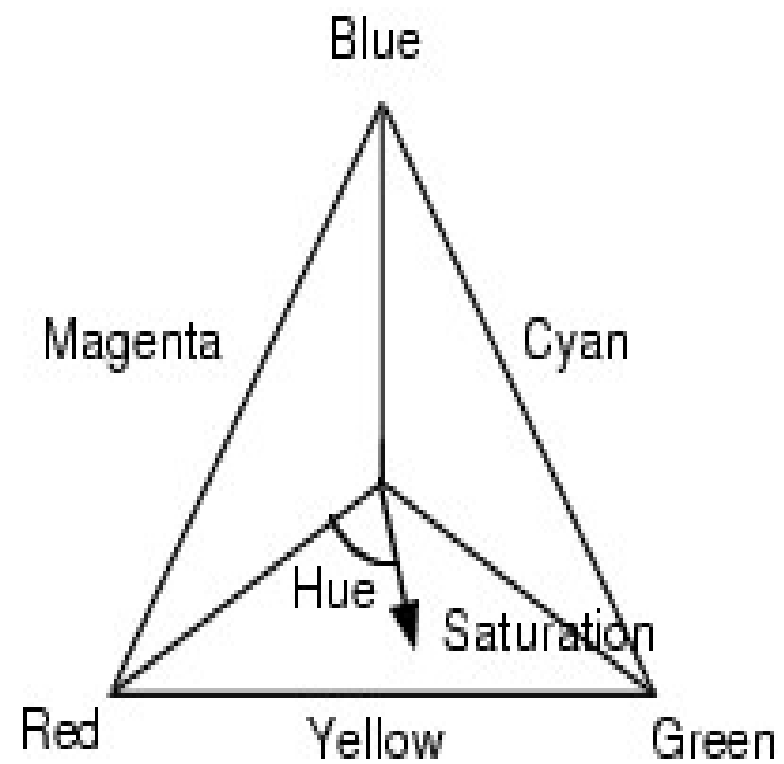
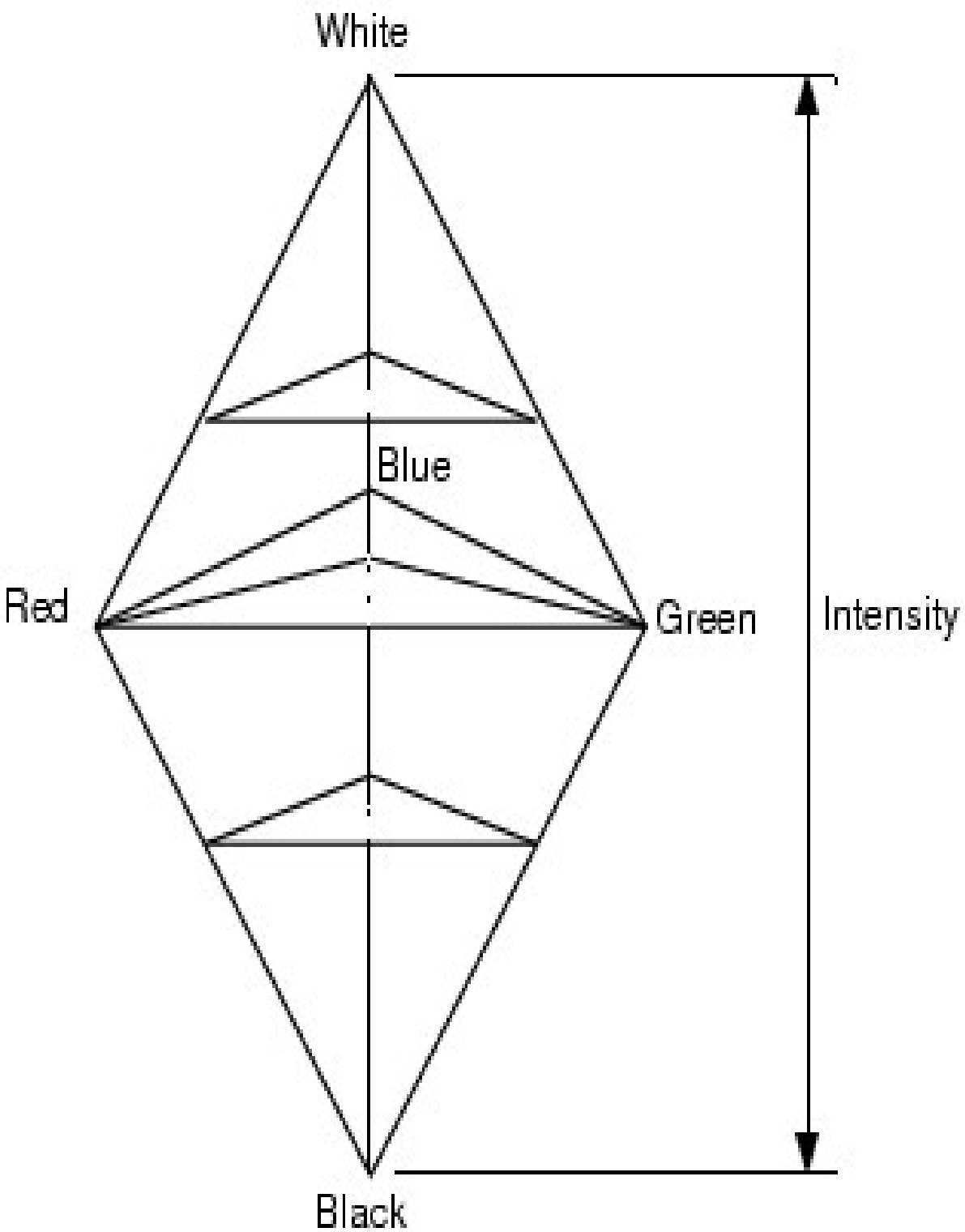
In practice, black is obtained by combining cyan, yellow and magenta, however, this leads to a muddy looking black. For publishing, the black colour plays an important role, therefore, in the CMYK colour model, black is added as the fourth colour, where K stands for black.

The HSI Model

This colour model is very close to human colour perception which uses the hue, saturation and intensity components of a colour, when we see a colour, we cannot describe it in terms of the amount of cyan, magenta and yellow that the colour contains.

Therefore, the HSI colour model was introduced to enable describing a colour by its hue, saturation and intensity/ brightness. **Hue** describes the pure colour, **saturation** describes the degree of purity of that colour while **intensity** describes the brightness or colour, sensation. In a grayscale image, intensity defines the gray level.

Fig. 6 shows the HSI colour model and the way colours may be specified by this colour model.



In Fig., the HSI colour model is represented and its relation to RGB model is shown in the Fig. (b). The HSI triangle in Fig. (b) shows a slice from the HSI solid at a particular intensity as shown in Fig. (a). You may notice that in Fig. , the hue, saturation and intensity values required to form the HSI colour space can be computed using the RGB values.

INTRODUCTION TO COMPUTER VISION

Computer vision is the field of study that endeavours to develop techniques to help machines understand the content of images and videos captured by single and/or multiple cameras.

It seems to be a trivial problem for human beings to understand and recognize contents of images and videos once seen, however, images and videos have a lot of content and it is not easy for a machine to focus on the relevant portions of an image, understand the content, recognize familiar objects and faces.

Applications of computer vision exist in every sphere of life, from medical image understanding to autonomous vehicles and robotics. Therefore, it is an important field of study.

CAMERA MODELS

A camera is a basic tool in computer vision. It is essential to record an image of the scene around us to be able to analyse the scene.

Therefore, it is important to learn the model of the camera. In general, a camera is a physical device that captures an image by allowing light to pass through a small hole (the aperture) onto a light sensitive surface.

The lens in a camera focus the light entering through the aperture onto the imaging surface or the lightsensitive surface. Also, a shutter mechanism controls the amount of time for which the photosensitivesurface is exposed to light.

THE PIN-HOLE CAMERA MODEL

In a pin-hole camera, a barrier with a small hole (pin-hole) is placed between the 3D object and the 2D image plane (light-sensitive plane).

Each point on the 3D object emits/ reflects multiple light rays out of which only one or very few of these light rays fall on the image plane by passing through the pin-hole.

Thereby, one can see that there exists a mapping between the points of the 3D object and the 2D plane forming an image and such a mapping will be a projection where the pin-hole is called the centre of projection.

Formulation of the pin-hole camera model

Assume that the centre of projection is the origin of the 3D Euclidean coordinate system. Let $Z = f$ be the image plane or the focal plane. Then, the centre of projection ('C') is called the camera centre or optical centre and f is the focal length.

The principal axis or the principal ray is the line from the camera centre that is perpendicular to the image plane. The point of intersection of the principal axis and the image plane is known as the principal point.

The principal plane/projection plane is the plane passing through the principal point parallel to the image plane, i.e., the image plane becomes the projection plane.

Under the pin-hole camera projection, a 3D point $X = (X, Y, Z)$ is mapped to a 2D point $x = (x, y)$ on the image plane where, the line joining the camera centre C to the 3D point X intersects the image plane at x .

From Figure 1, it can be seen that by rules of similar triangles,

$x = fx/z$ and $y = fy/z$.

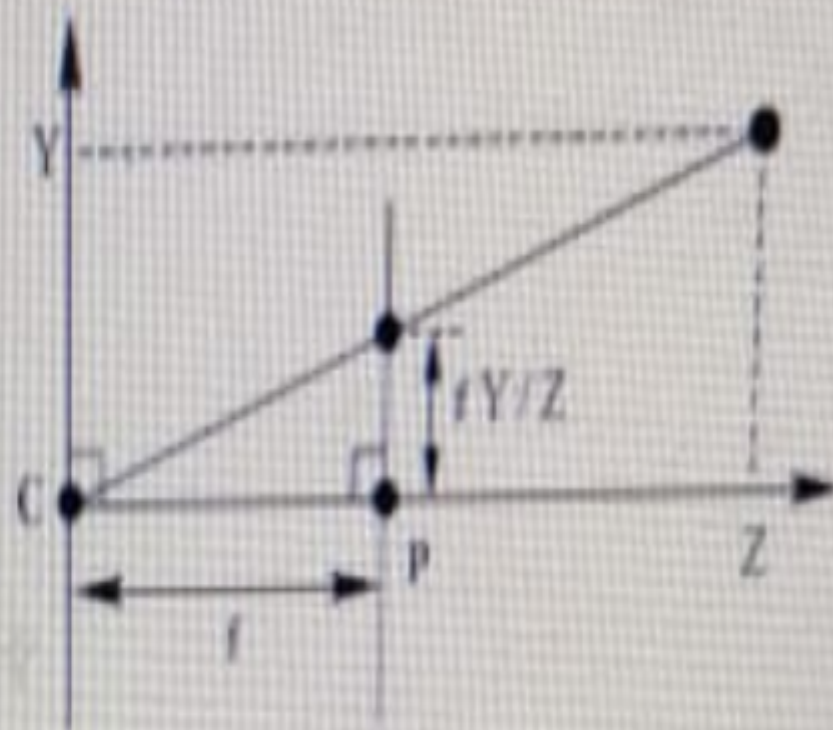
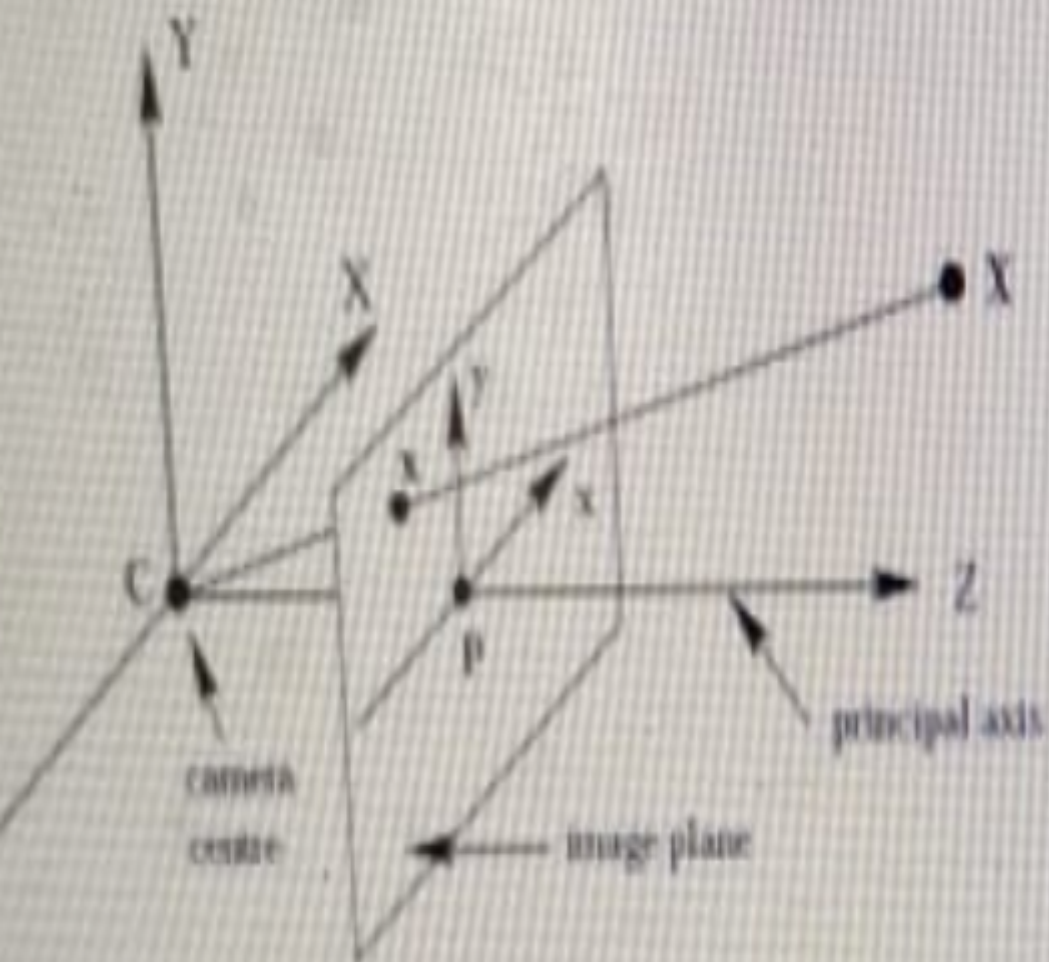


Figure 1. The pin-hole camera geometry. Figure taken from [1]

PROJECTIONS

Perspective projection

Consider the pin-hole camera model as a perspective projection transformation discussed in 8.4. Using matrix transformation in homogeneous coordinates.

the projection point in the Euclidean space can be obtained $X = f_x/z$ and $y = f_y/z$.

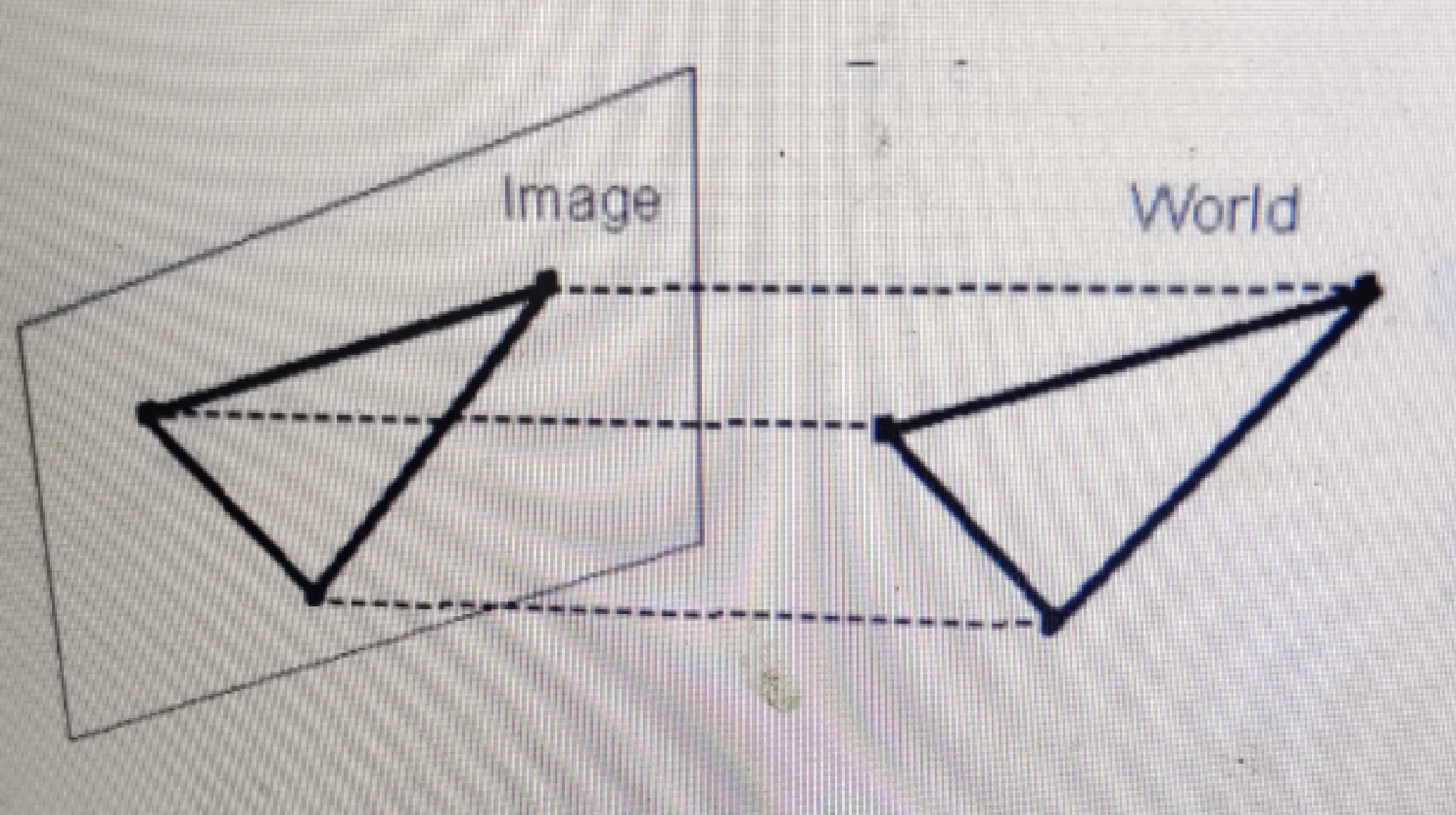
This way the perspective projection of a 3D point via the center of projection (the pin-hole) onto a point on the image plane $Z = f$ can be obtained in matrix form.

Orthographic projection

Orthographic projection is the projection of a 3D object onto a plane by a set of parallel rays that are orthogonal to the image plane, i.e., it is a parallel projection.

In this projection, the center of projection is taken at infinity. For any point (X, Y, Z) from the object, its orthographic projection on the image plane $Z = f$ is given by $x = X, y = Y$

The important properties of the orthographic projection are that parallel lines remain parallel and the size of the object does not change.



Orthographic projection is shown in the above image

Weak perspective projection

A perspective projection is a non-linear transformation, however, under certain circumstances, we can approximate it by a linear transformation, that is, as a scaled orthographic projection.

The important conditions for these are that the object lies close to the optical axis and that the dimension of the object are small compared to its average distance Z' from the camera.

Equations of the weak perspective projection are:

$$X = fx/z' \text{ and } y = fy/z'$$

in which each point is scaled by f/z .